

Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 4
Introduction to Layout and Post-Extraction simulation

Due: 02/19/2025

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1 Introduction

This exercise will introduce how to make layouts, use the Design Rule Check (DRC) and Layout vs. Schematic (LVS) tests, and run parasitic resistance and capacitance (RC) extraction (RCX). Students will design and verify the photo masks of an inverter and simulate its extracted results.

Check section 7 for the lab report requirement and section 8 for the required items for signoff. Please submit your lab report (with the signoff sheet attached) to myCourses

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

```
/classes/ee620/bin/unlock_cadence.csh
```

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

```
Options → Bindkey
```

Select a category (e.g. **layout**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!
- DO NOT delete any Cadence files from command line or file manager!

2 Prelab - Inverter Stick Diagram

Before coming to the lab, draw the stick diagram of an inverter layout. The stick diagram must be colored, properly labeled, and made using computer software (**not hand-drawn**). Be prepared to answer questions regarding the stick diagram.

3 Layout vs. Physical Device

When creating a layout, it is important to know what your design implies in the context of a physical device. While designing we only have control over two of the three dimensions (length and width) of the physical device. Control of the third dimension is reserved for microelectronic engineers and the fabrication process. Figure 1 shows the cross section of an inverter. This is an approximate representation of what it would actually look like after fabrication. **Be sure to compare your finished inverter layout with Fig. 1.** You should have a good understanding of how each layer in your layout corresponds to a fabrication step.

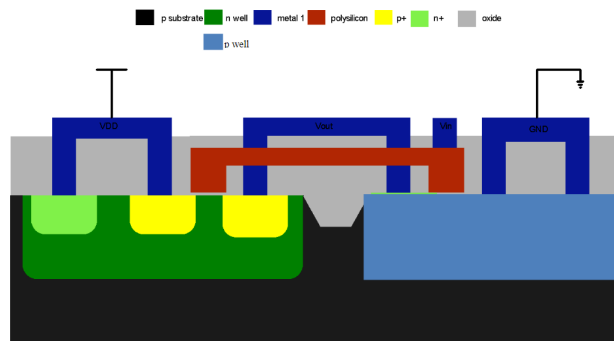


Figure 1: Cross Section of an Inverter

4 Procedure

Before starting this exercise, draw the stick diagram for the inverter schematic. When starting this lab, make sure you are on a fresh logged in session. If not, then close Cadence, log out and log back in. Open the Cadence tools and create the layout view for the inverter. From the Library Manager window, select your library, then your inverter cell. Then select **File → New → From Cellview**

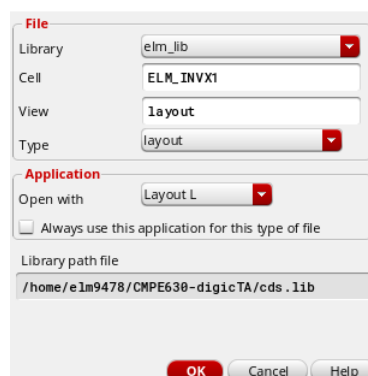


Figure 2: Create New Layout Cellview Form

Click OK.

A warning window may appear indicating that the license for Layout L is not available, and do you wish to use Layout XL instead. Click Always and a new window pops up.

The layout editor opens as shown below. On the top are text based menus, and icons for the most often used commands.

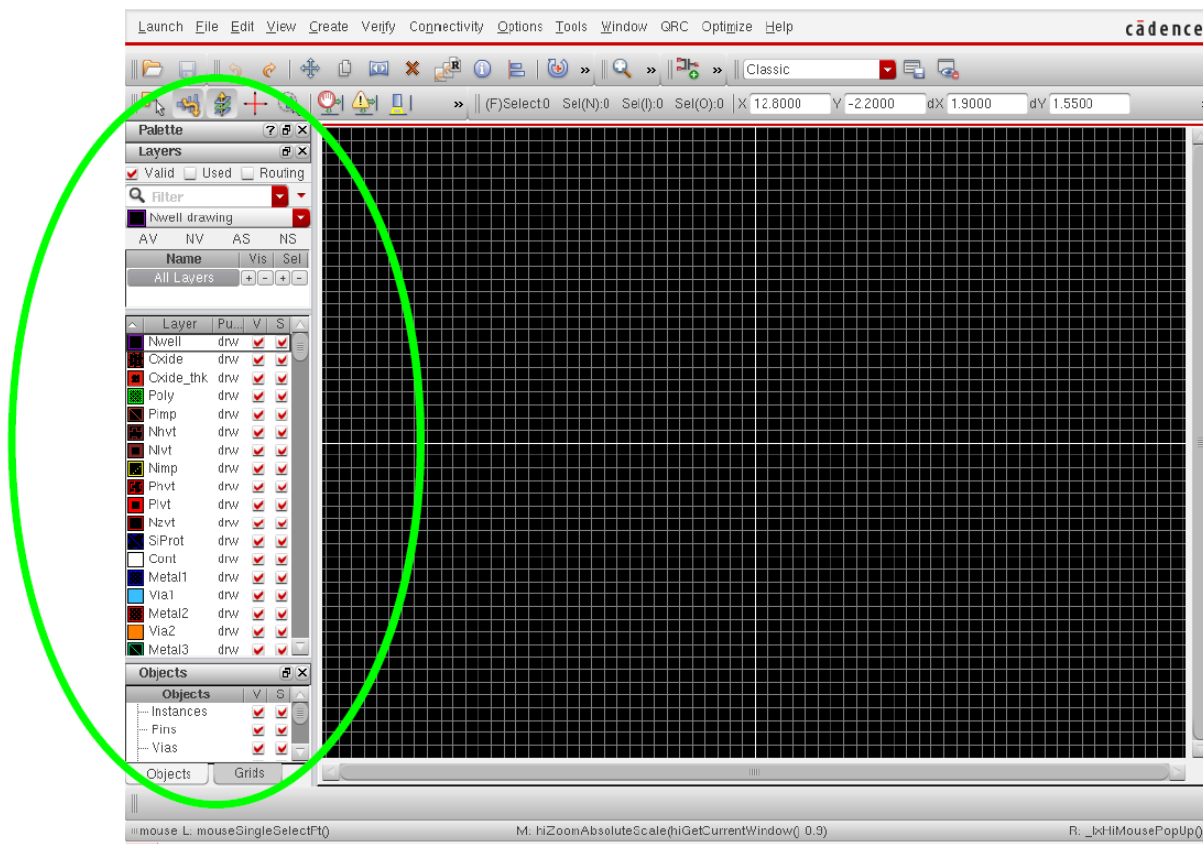


Figure 3: Virtuoso Layout Palette Panel

You may find it easier to enlarge the layout window. As in schematic capture, at the top you will find all text based menus and icons representing the most often used commands.

Note that one important panel to understand is the Palette on the left (Circled in Figure 3). The palette presents a concise view of the PDK defined layers and allows you to control which layers are visible (V checked), selectable (S is checked), and active (the layer highlighted). The palette also allows you to control which objects and grid items are visible (V is checked), and selectable (S is checked).

Before we proceed, we will need to check a few settings. Click **Options → Display** and adjust your Display as necessary to match the form shown in Figure 4

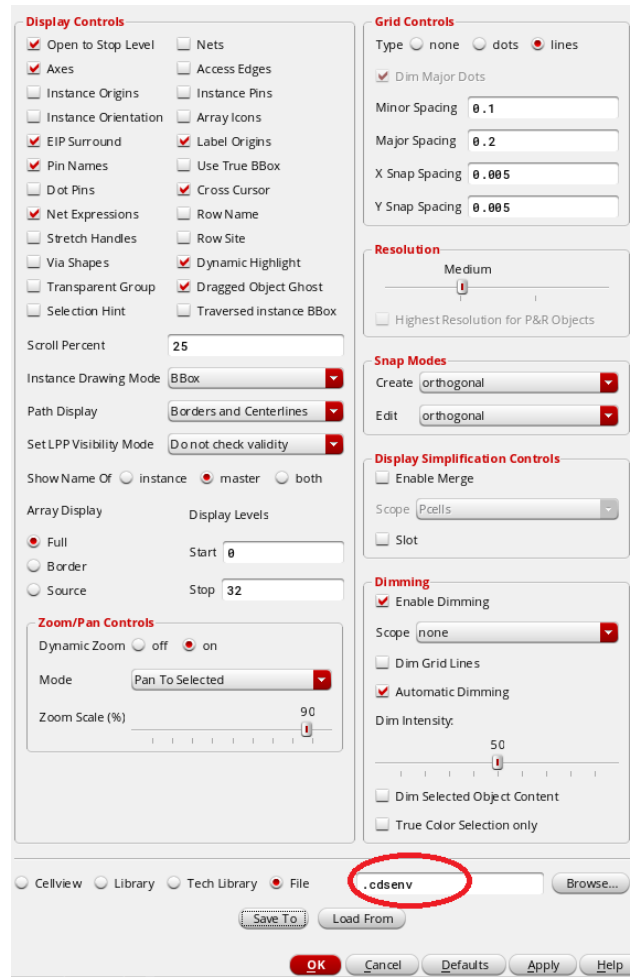


Figure 4: Display Options

Your environment should be already configured, however, if necessary (which is almost never) edit the field in Grid Controls, particularly Minor, Major, X and Y Snap Spacing.

CAUTION: You should **ONLY** modify X and Y Snap spacing before creating layout views. Changing the X and Y Snap Spacing after creating layout views can (WILL) cause your cells to be OFF GRID, causing DRC violations and making the layout **UNUSABLE**. **In this case you will be forced to delete your layout and start over.** Always get in the habit of checking X and Y Snap Spacing Display Options to make sure you are using the proper spacing.

Then click the file button toward the bottom of the form, and then click Save To to save your setting to the .cdsenv file as shown above. **Note that the file name is ".cdsenv" NOT ".cdsenv".** Then click Apply; then click OK to return to the layout window.

4.1 Layout Process In Detail

We will now start the layout of a standard cell; for most standard cell designs the height will be fixed. Note that the total width is not enforced but strictly follows the routing grid pattern. To

enable better design practices, we use a parameterized layout template for our cells that include the n-well region (the bulk of the pmos transistors), p-well region (the bulk of the nmos transistors), and well ties.

Select **Create → Instance** and browse to select the cell template as shown below (see Figure 5).

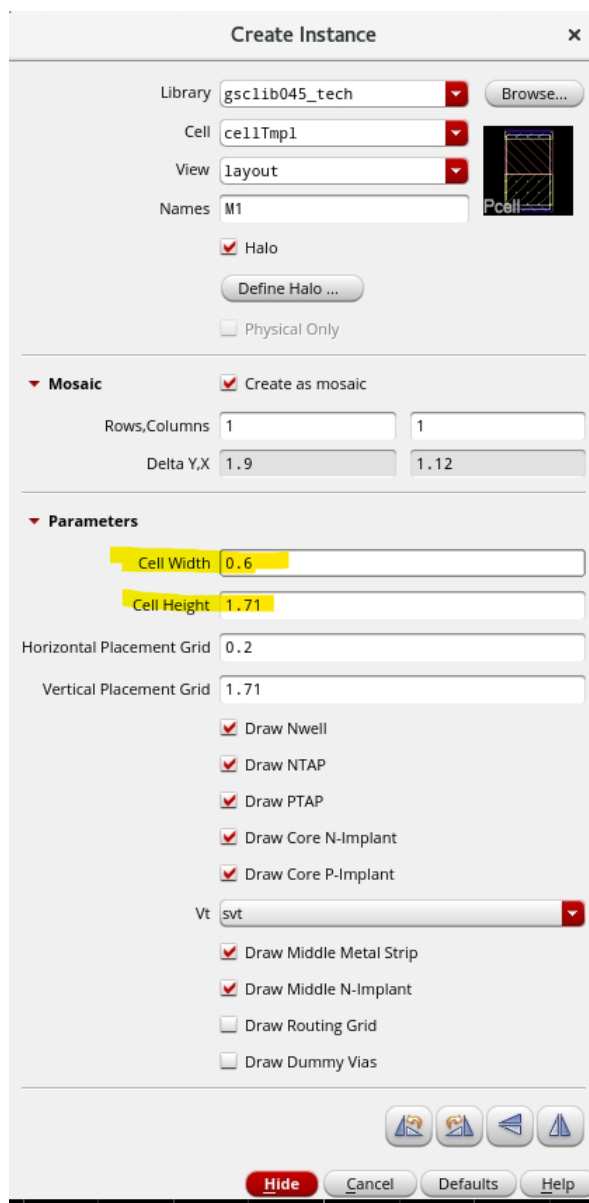


Figure 5: Create Cell Template Layout Instance

Adjust the Cell Width to **3 tracks or $24\lambda(0.6\mu m = 24\lambda)$** ; the cell height is fixed at $1.71\mu m$ which is a bit more than 8 tracks to account for the power straps and well ties. Click **Hide** and place the cell template at the origin (0,0) as shown in Figure 6. Note that you may wish to temporarily change the grid display from lines to dots to make it easier to place the instance at the origin.

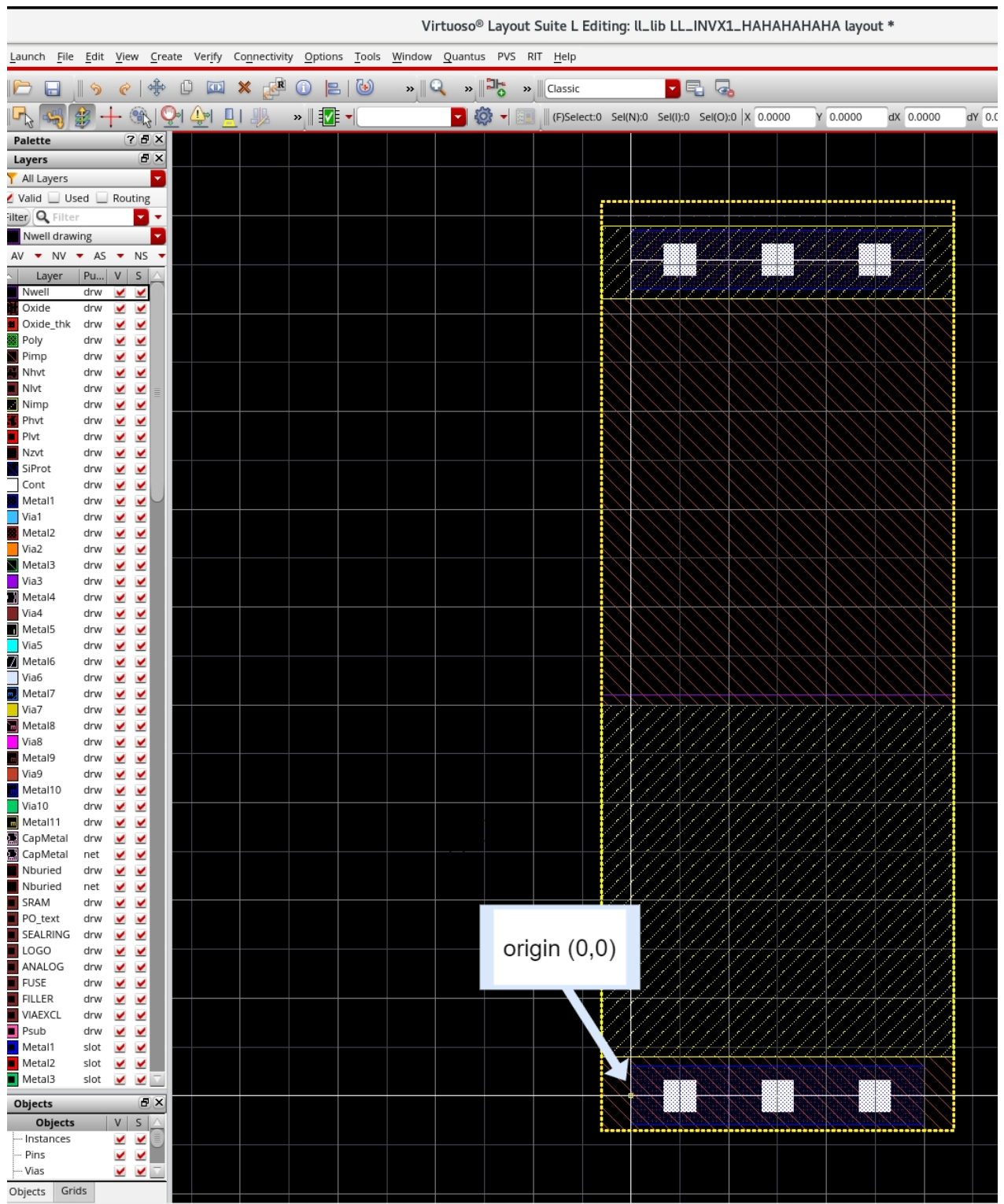


Figure 6: Cell Template Placed In Layout

Move the edge to the desired position and click again to place it. Before we start drawing, we will orient our thinking by showing you a finished inverter layout in Figure 7.

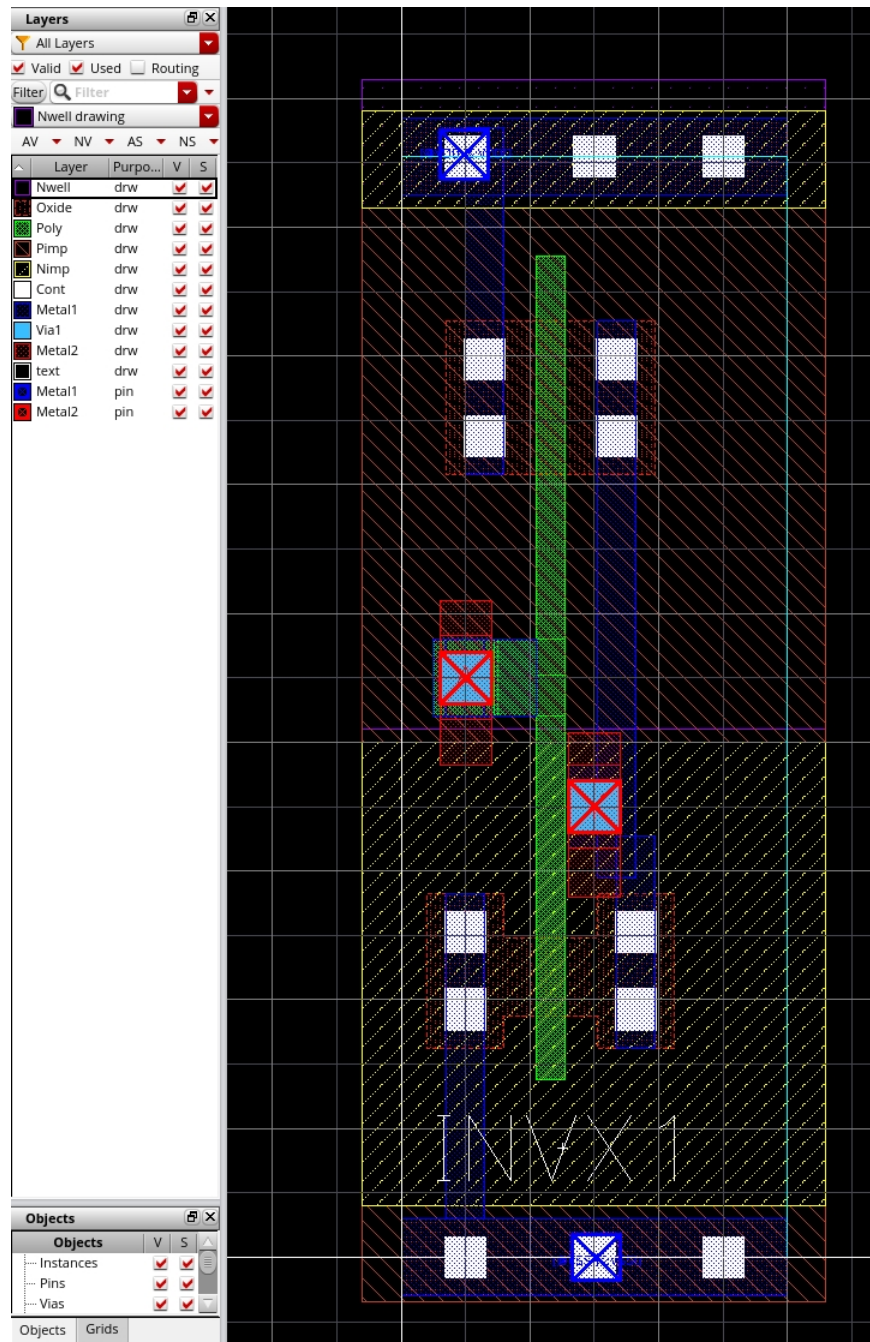


Figure 7: Example Inverter Layout Reference

You will re-create this inverter layout for the first part of this lab exercise. But before you get started, please **read the following very carefully**.

You will draw polygons of the FOLLOWING LAYERS for all standard cell layouts:

- **Oxide:** This is used to define the transistor gate (the presence of oxide under the polysilicon gate forms a channel and what is known as the gate capacitor), in other regions, the oxide is stripped away to define the diffusion regions which form the transistor source and drain. You should strive to make oxide diffusion regions as small as possible to minimize parasitic

capacitance, but large enough to carry the current necessary without causing a significant voltage drop.

- **Poly:** Polysilicon can be used for routing, as it only forms a channel and is implicitly a transistor over an oxide diffusion region.
- **Metal1 and Metal2:** Metal layers are used for routing. However, we will **only use Metal1 for routing, as Metal2 will only be used for I/O pins**
- **Contact:** This is used as a via to connect Metal1 to oxide and poly layers

Be aware of the following when you draw your polygons:

- There are a lot of different layers available in the Layout Palette Panel. We will only use the layers mentioned above
 - You will see layers like Pimp, Nimp, and Nwell that are also used in your layout. Do not remove them or modify the size of them. Those are created from the cell template and are used for defining the body connections of the transistors.
 - Also, be aware that there are metal layers with the same name but they are used for different purposes. For instance, you will find multiple Metal1 layers in the palette. **Make sure to use Metal1 drawing for routing**
- This technology/process has different sizing requirements for different layers. Details for that can be found in the gpdk045 reference manual. Alternatively, have active DRC check  on. This tool will inform you when a polygon does not meet the sizing requirement or spacing requirement between layers.
- Useful shortcuts for creating your design (There are more you will learn on your own)
 - Rectangle (R) - Creates a rectangle based on two clicked points on the design
 - Undo (U)
 - Redo (Shift + U)
 - Crop (Ctrl + C) - Cuts sections of a polygon
 - Move (M) - Allows for movement of polygons based on cursor movement
 - Stretch (S) - Extends the selected side of a rectangle based on cursor movement
There are various other commands such as Move (Edit → Move) to move a polygon, etc.
 - Ruler (K)

Take a moment to review the menus and better understand the hotkeys, as these commands will be used frequently, and learning the shortcuts will greatly expedite your workflow.

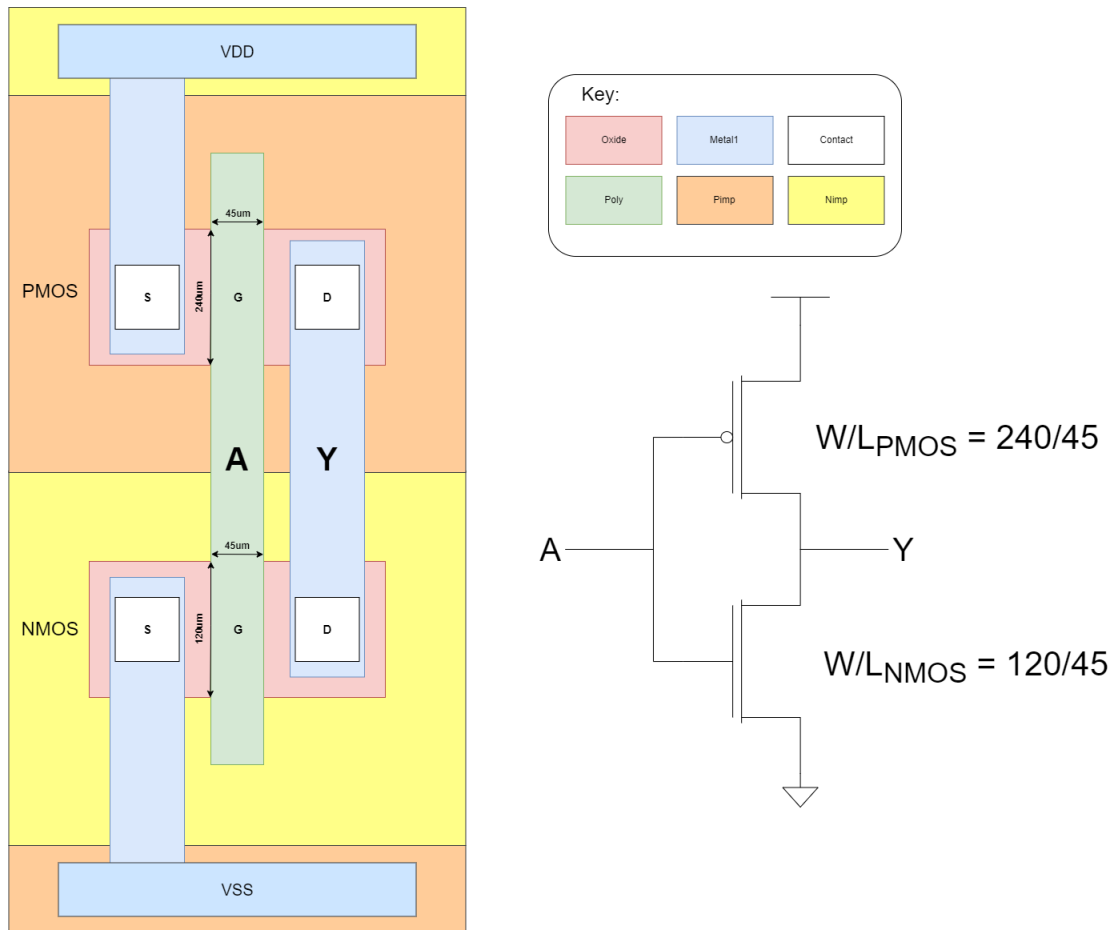


Figure 8: Comparison Between Layout and Schematic

Figure 8 shows the mapping of each layer in the layout to the schematic. Below are a few key highlights:

- The width of your poly is the **length** of your transistor, as it is also the distance from drain to source
- The length of the poly that overlaps with the oxide is the **width** of the transistor. See how only 240nm of poly overlaps with the oxide layer on the PMOS side
- Pimp and Nimp layers are a part of the cell template. You will not create polygons for those layers
- The white squares are contacts. Although it is not shown in Figure 8, you need **at least 2 contacts** to connect Metal1 to the oxide region to ensure strong signal transmission. The source and drain of the transistors are formed based on the diffusion across the gate
 - Metal1 is used for connections involved the source and drain of the transistor



Before you move on, make sure to have fully read the previous pages. We do not expect you to fully understand all of the rules and advice, but please make sure to be aware of them and refer to them as you work on your layout. Now is also a good time to ask the TAs questions!

With that said, you will first start by creating the layout **without any pins**. We will place the pins after we have finished drawing all of the polygons. To give you some help, we have hidden the pins from the layout, which is shown in Figure 9.

Additionally, see Figure 10 and 11 for the dimension of some of the polygons used.

Before you start...

- From the layers panel, choose the layer drawing you want to create a polygon of. Then press **R** to draw a rectangle
- Press **K** for the ruler, the ruler's dimension is in um
- Undo is **U** and Redo is **Shift+U**
- Make sure to have DRC check  on
 - There are a lot of DRC rules stating the minimum distance between polygons, the minimum size of a polygon, etc. DRC check will display live DRC errors (if there are any) as you are moving/creating polygons.

Now, without further ado, try to replicate the layout that is shown in Figure 9.

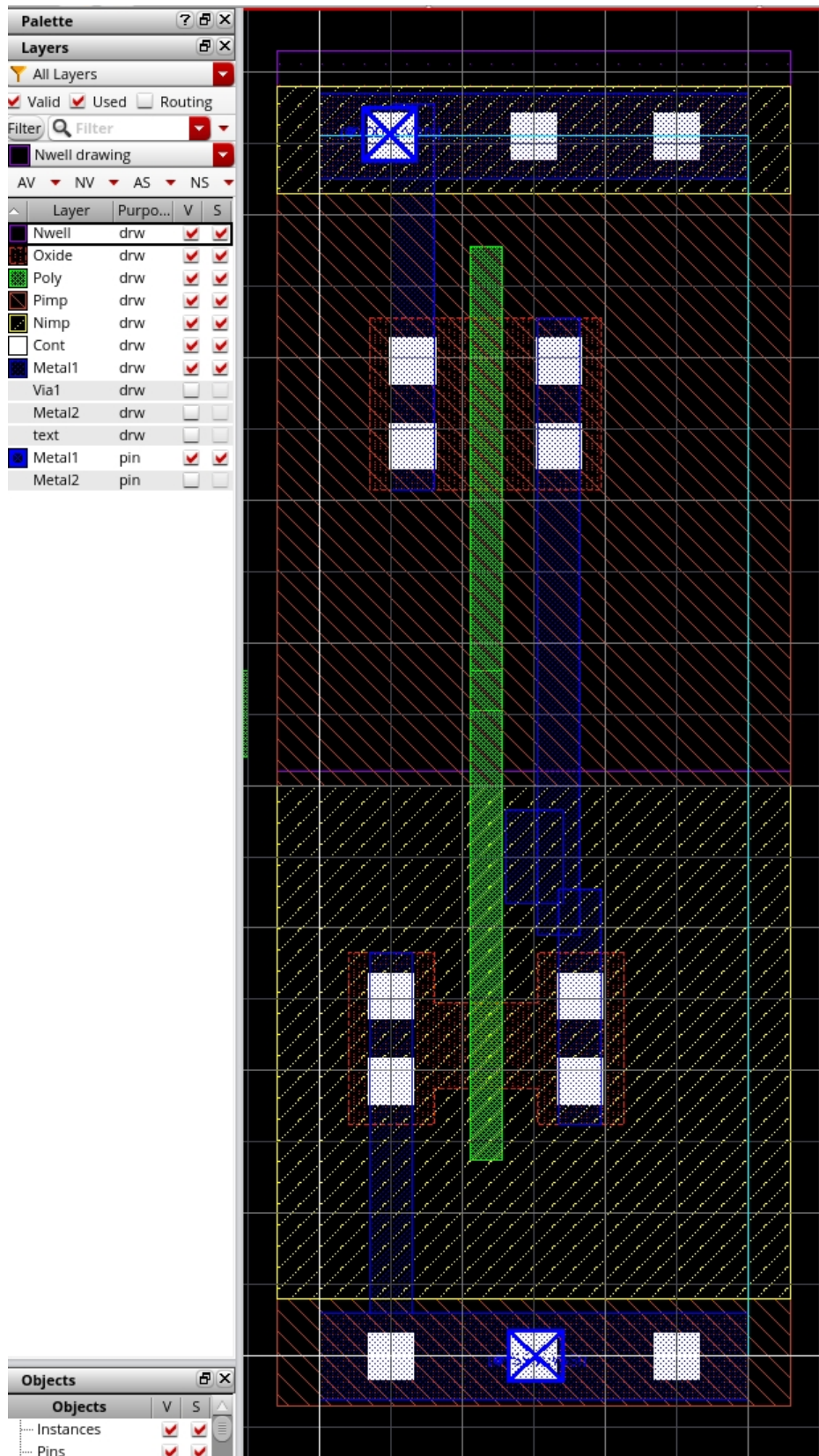


Figure 9: Inverter layout without I/O pins

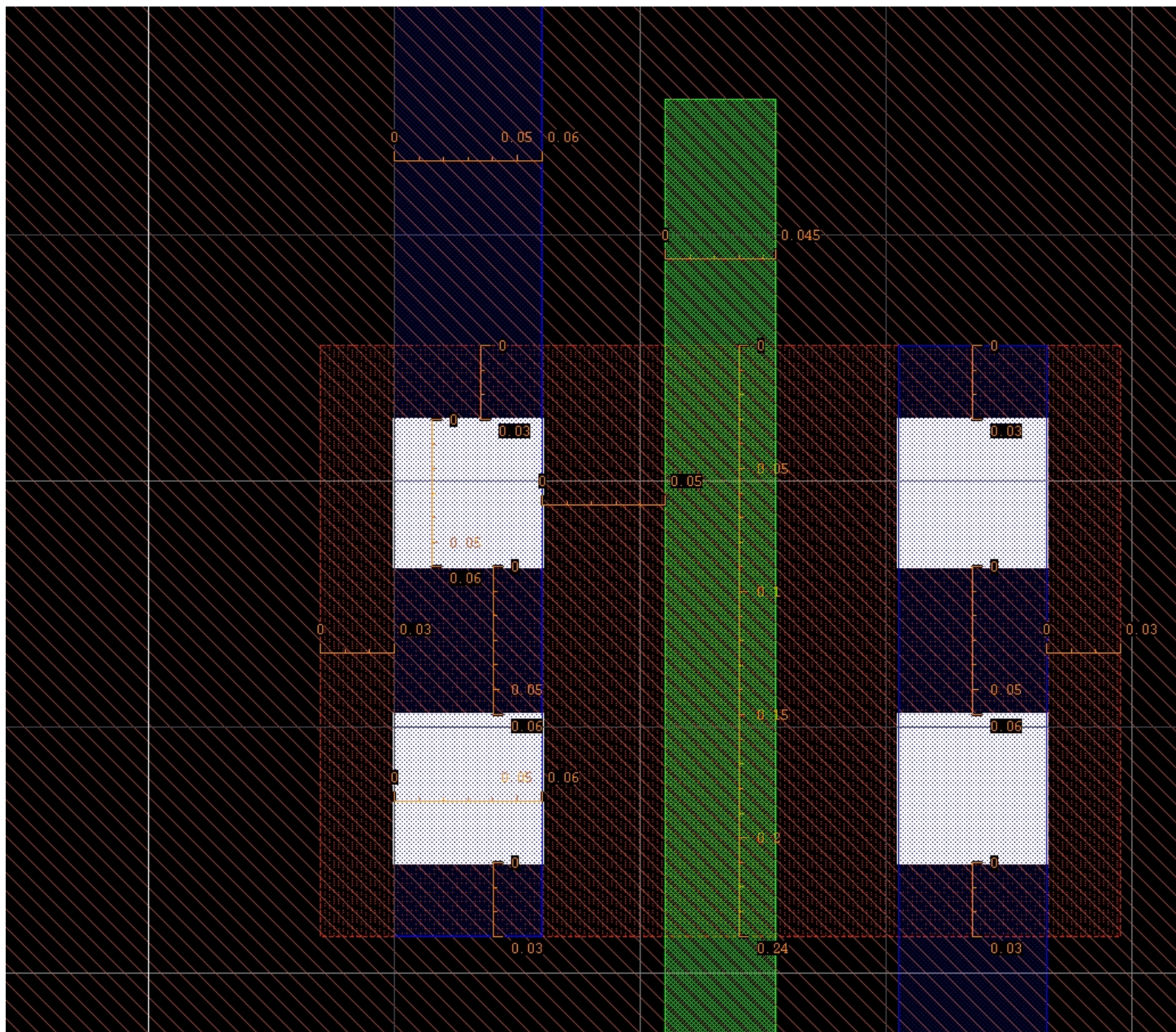


Figure 10: Dimension of the PMOS layout

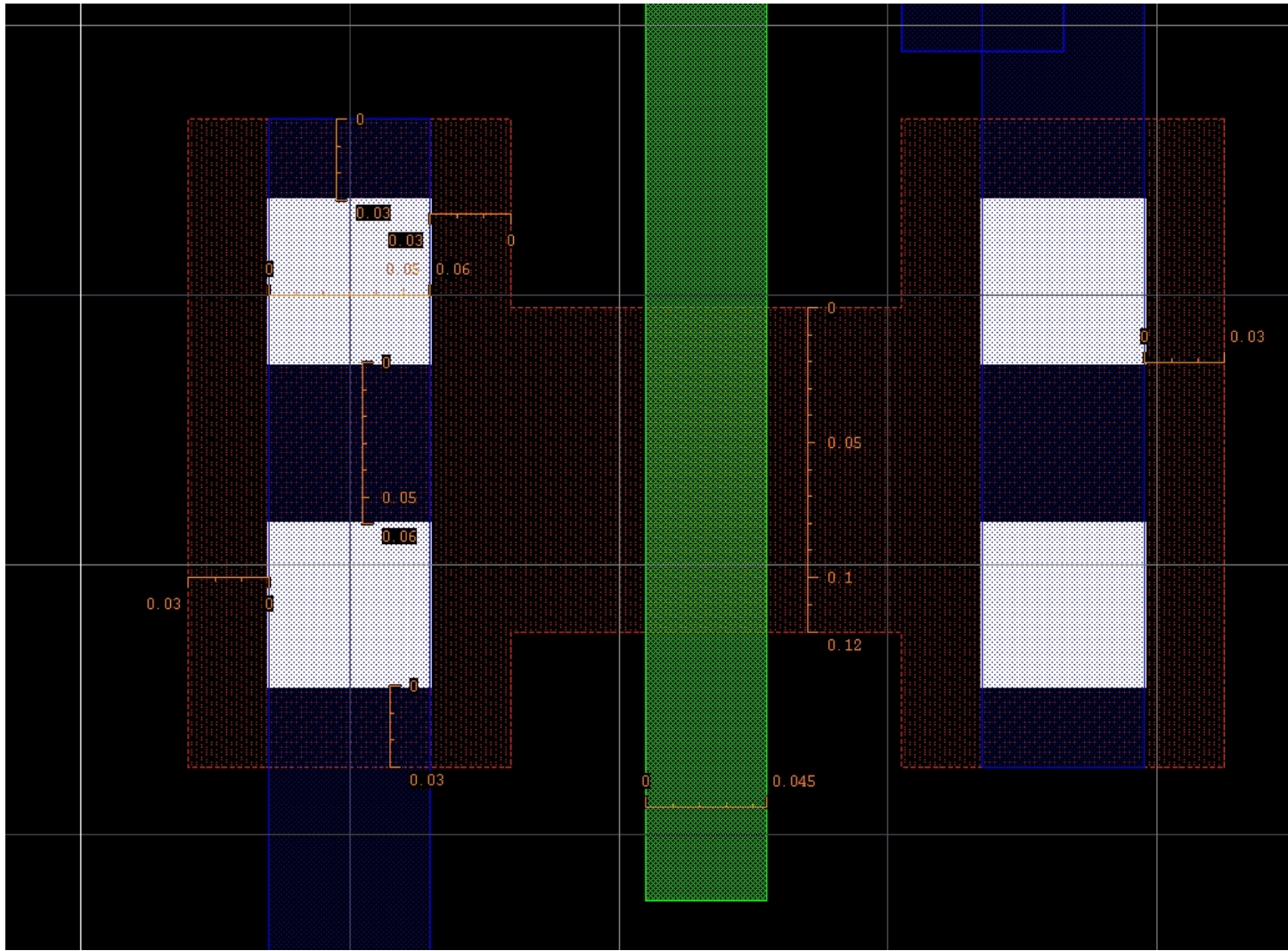


Figure 11: Dimension of the NMOS layout

At this point, only pins, text labels, and PR boundary box are missing from our layouts. First, we will add the pins. Pins are used to connect each cell using the place and route tool (which you will use in the future) and are the ports of your instance. Let's start by creating the input pin for the signal 'A'.

The first step is to select the layer *Metal2|Pin*, make sure this is the active layer. Note that pins are $0.08\mu m * 0.08\mu m$ in size.

Now, before you start placing pins, please be mindful of the following:

- These pins **must be staggered on 1/2 tracks (at the center of two grey lines)** to minimize DRC violations when creating more complex layouts using these cells in the future. I/O Pins (and labels) need to be on the **Metal2— Pin** layer only. *You will need to draw a **vertical** metal 2 strip surrounding the metal2 pin for future auto routing. Ensure that the area of metal2 is at least $0.02\mu m^2$ otherwise, there will be a DRC error¹. Each row and each column can only*

¹Virtuoso uses μm as the unit for area

have one I/O pin.

- Take a look at Figure 14, which shows the placed I/O pin along with other polygons needed for passing DRC. Did you notice the light blue polygon underneath the I/O pins? Those are called vias, and we use them to connect objects of different layers.
 - For the output pin Y, you will create a via connecting Metal1 to Metal2 (drw layers) underneath the pin as shown in Figure 14
 - To create a via, select Create → Via (or press O) and the following form should pop up (see Figure 12). By default the via definition is set to **M1_PO**, which is a Metal1 to Poly via. You can change it to a via of another definition with the drop-down menu. For instance, **M2_M1** creates a via from Metal2 to Metal1
 - For the input pin A, you will create a connection from Metal2 → Metal1 → poly. This can be performed by placing two vias, one connecting Poly to Metal1 (M1_PO), and the other one connecting Metal1 to Metal2 (M2_M1)



Figure 12: Create Via form

Now, to make the actual pin , select **Create → Pin** and the following form appears.

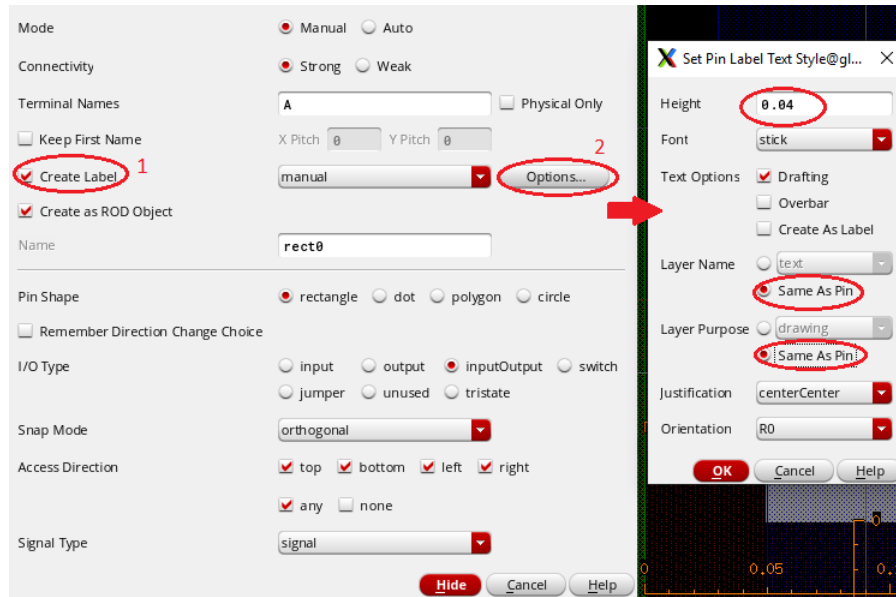


Figure 13: Create Layout Pin Forms

Again, Figure 14 shows the placed pin along with via and the needed polygons surrounding the pins (please seek for helps from TA if there is any confusion here).

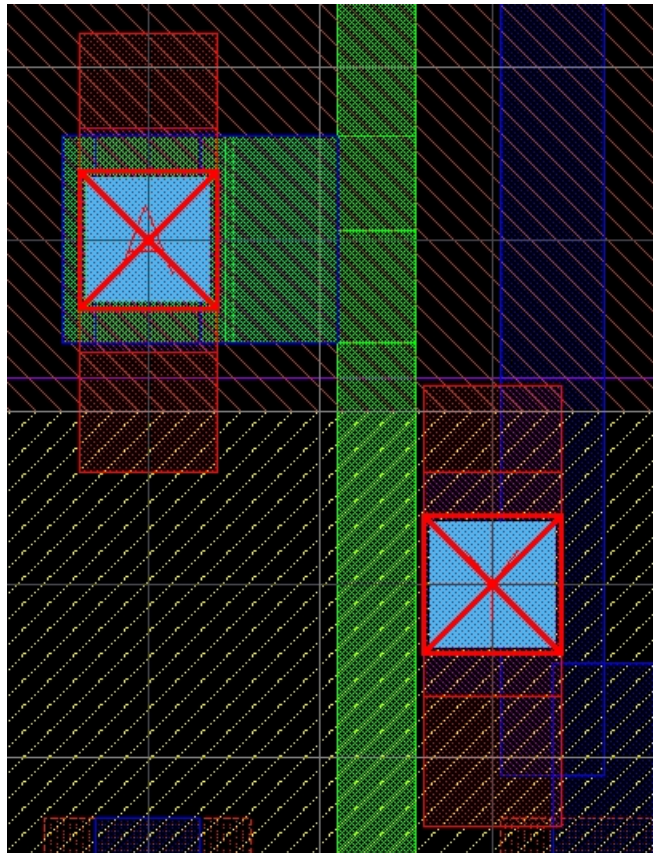


Figure 14: I/O pin placement

After you have created the I/O pins, it is time to make the VDD and VSS pins, which is very similar. Let us create and place the VDD pin first.

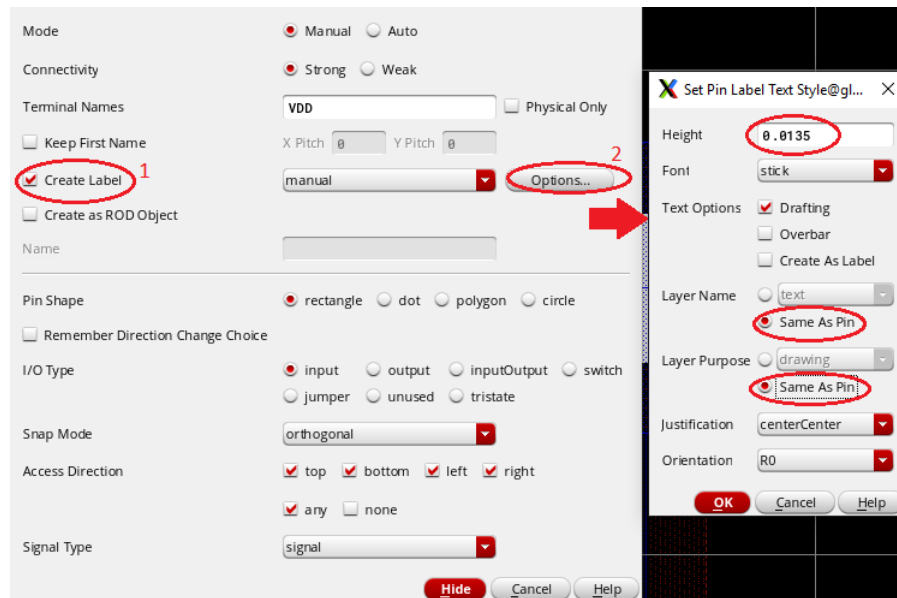


Figure 15: Create VDD Layout Pin Forms

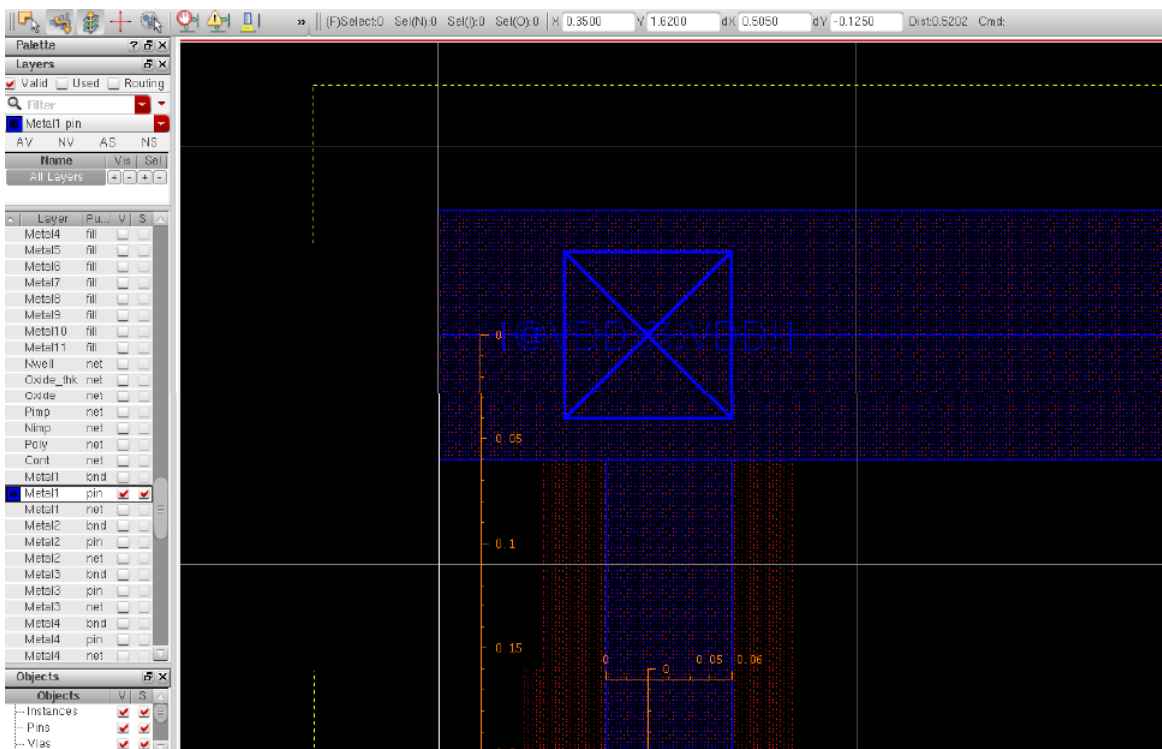


Figure 16: Placed Pin VDD

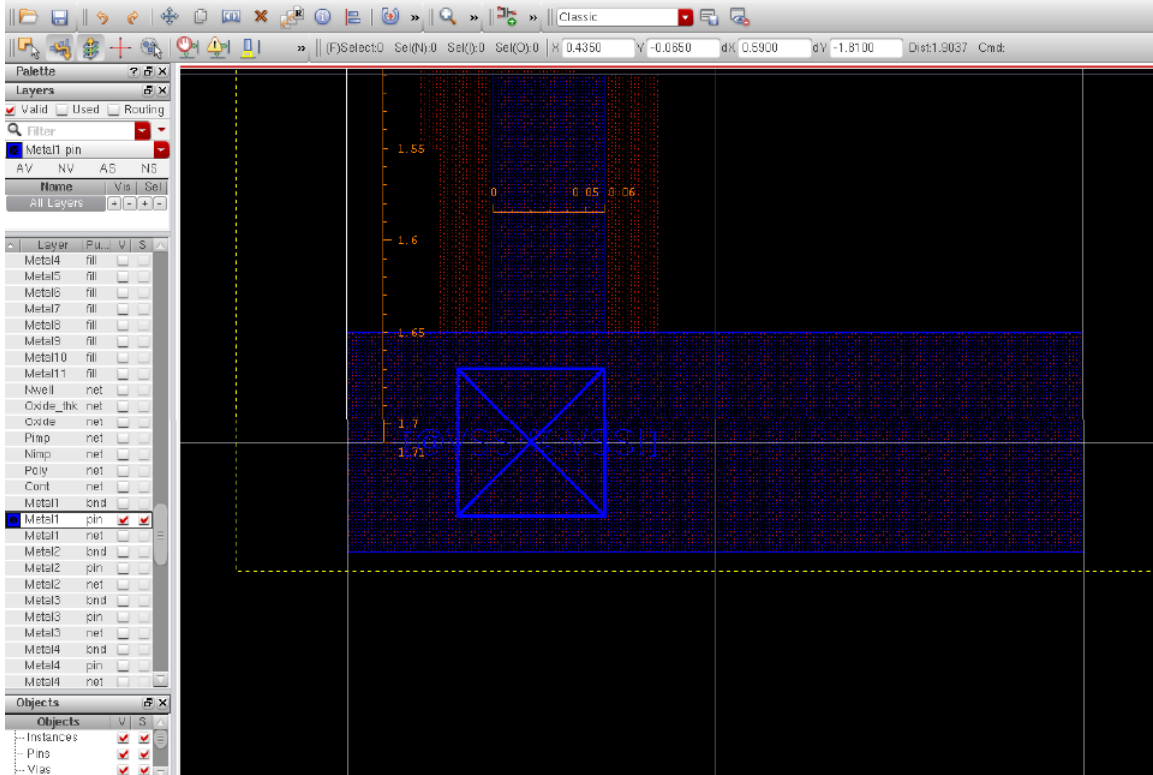


Figure 17: Placed Pin VSS

VDD/VSS Pins (and labels) need to be on **Metal 1— Pin only**. **VDD will be at the top and VSS will be at the bottom of the cell** as shown in the reference diagram (Figure 16 and 17). For labeling purposes, recall the special treatment of VDD/VSS as these are **global signals VDD! and VSS!**. The VDD and VSS pins do not need to follow the same placement rules as the I/O pins since they are global signals. You should check the properties of these pins. The following windows show the pin properties for the VDD pin.

Figure 18: Edit Layout Pin Properties Form - Attributes

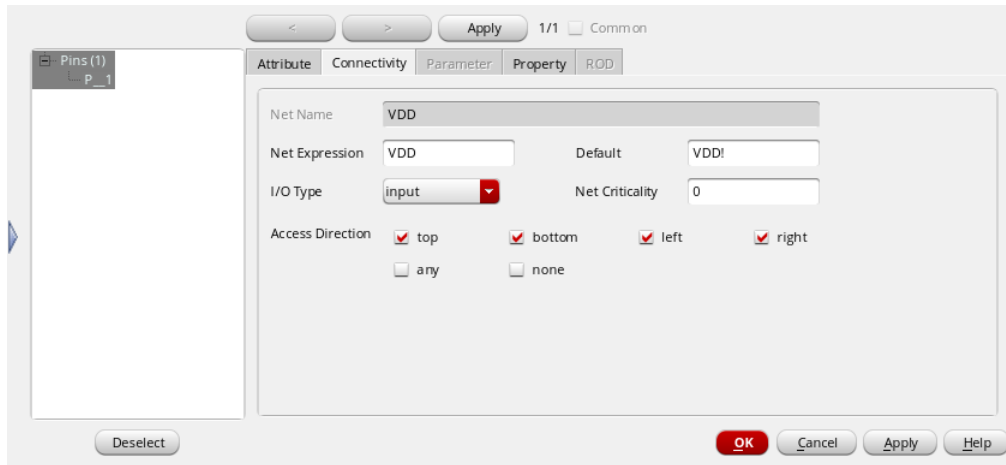


Figure 19: Edit Layout Pin Properties Form - Connectivity

Check that the VSS pin is configured similarly, use P_2 for the Pin name.

Your layout is almost complete. We have two final items to create: (i) a Place and Route (PR) boundary, and (ii) a text label to identify our cell.

Select **Create → P&R Objects → P & R Boundary** as shown.

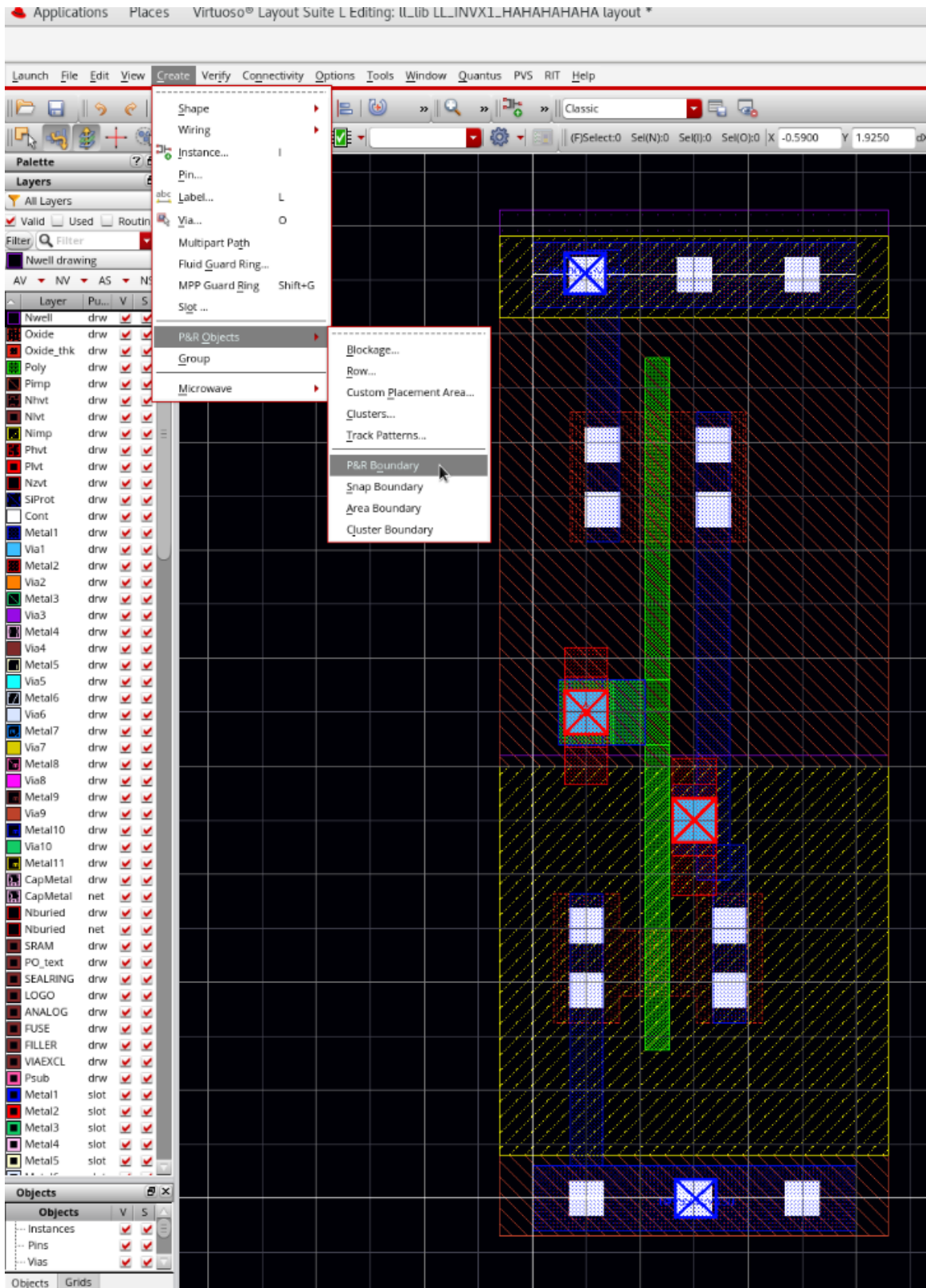


Figure 20: Create Place and Route

Now draw a rectangle over the inner region of the cell (Over the existing white box within the cell which was used for placement at the origin). Your P&R Boundary should appear as in Figure 21 and should have dimensions of $1.71\mu m * 0.6\mu m$

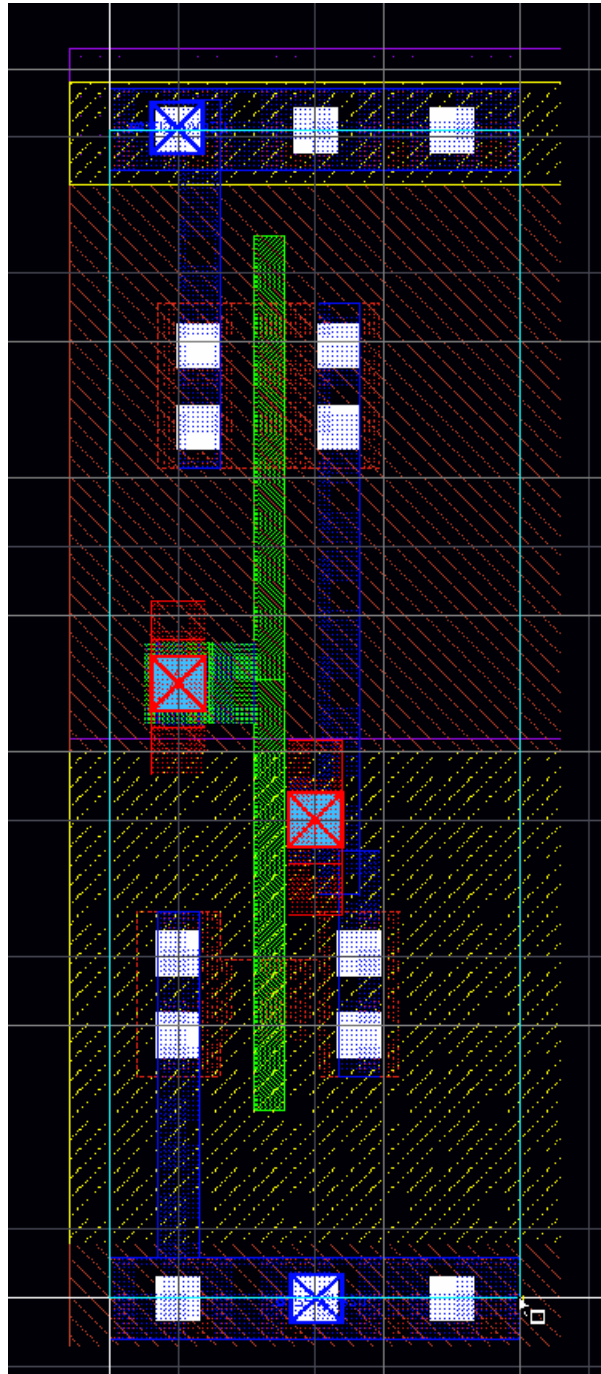


Figure 21: P&R Boundary Placed

Now let's label our cell. Select Create → Label and the following form appears.

Mode ☒ Manual ☐ Auto

Label (Pattern)

Label Layer/Purpose

☐ Use current entry layer
☐ Use same layer as shape, select purpose
☐ Auto (Using stampLabelLayers section from technology file)
☒ Select layer

Label Options

Font Height Justification
☒ Drafting ☒ Attach
☐ Overbar ☒ Expand
☐ Keep Label

Scan Line

☐ Label Selected Objects ☒ Auto Zoom Out

Auto Step

☐ Step X Y

Snap Mode

Figure 22: Create Layout Label Form

Click Hide and place the label as shown in Figure 23

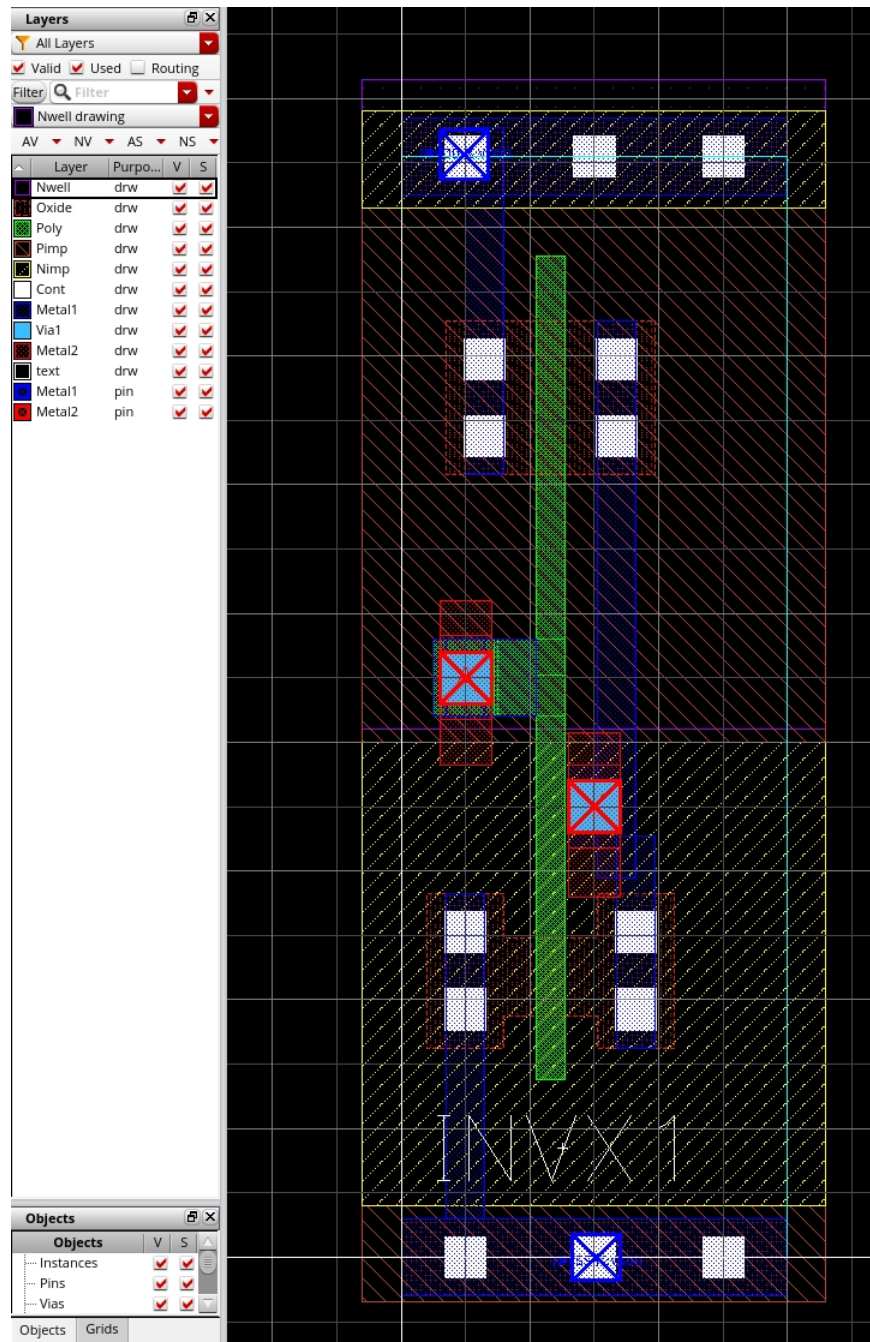


Figure 23: Final Inverter Layout with Text Label

One final step is to identify the cell site to allow the automatic placement tools to build rows out of our cells. **Select File → Properties** to open the file properties form.

The screenshot shows a software window titled 'Layout File Properties Form'. It has a tabbed interface with 'Attribute', 'Connectivity', 'Parameter', 'Property' (selected), and 'ROD' tabs. The 'Property' tab contains several input fields and dropdown menus. The 'Library' field is set to 'elm_lib', 'Cell' to 'ELM_INVX1', and 'View' to 'layout'. The 'bBox' field shows the coordinates '((-0.06 -0.07) (0.46 1.83))'. The 'Type' is 'maskLayout', 'DBU per micron' is '2000', 'Mode' is 'Edit', and 'Save Needed' is 'Yes'. The 'CellType' dropdown is set to 'core', and 'Symmetry Type' is set to 'no'. A 'SitePattern' section is expanded, showing 'Site Definition' as 'CoreSite' and 'Rotation' as 'Unknown'. Below this, 'Origin: x' and 'y' are both set to '0'. At the bottom of the window are four buttons: 'OK', 'Cancel', 'Apply', and 'Help'.

Figure 24: Layout File Properties Form

Select the **Site Definition CoreSite** under the SitePattern area of the form. The file properties Site Definition should look like the Figure 24. Click OK and then save and close your inverter layout.

And now the real fun starts.

Up to this point we have not run a Design Rule Check (DRC). Since we ran through an example you will have to check for errors.

Disclaimer:

- The figures for the remaining instructions show a different inverter layout. Do not worry about that as the usages of DRC, LVS, and QRC remain the same. Also, this provides another example of an inverter layout to compare yours to. In many cases, this layout provides poorer performance than the one we just created. As an interesting self-exercise, see if you can figure out why this may be the case
- The figures for the remaining instructions show a slight different GUI for the DRC, LVS, and QRC tools. Do not worry about that as the settings are mostly the same

4.2 Run Design Rule Check (DRC)

We will be using the integrated Physical Verification System (PVS).

Then logon, return to your cmpe630 directory and **restart** the Cadence tools and open your Inverter layout.

PVS should be enabled by default, if not, select **Launch → Plugins → PVS** as shown below (see Figure 25).

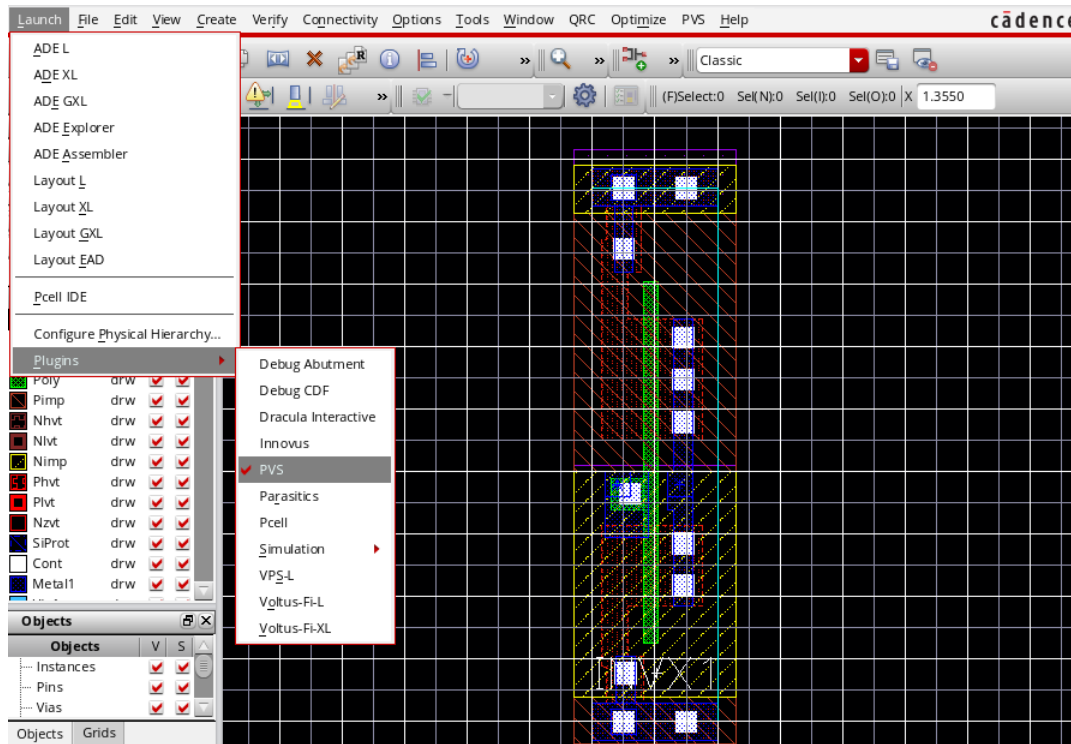


Figure 25: Layout Launch PVS

You will now see a PVS option appear on the top menu. Select **PVS → DRC** as shown in Figure 26.

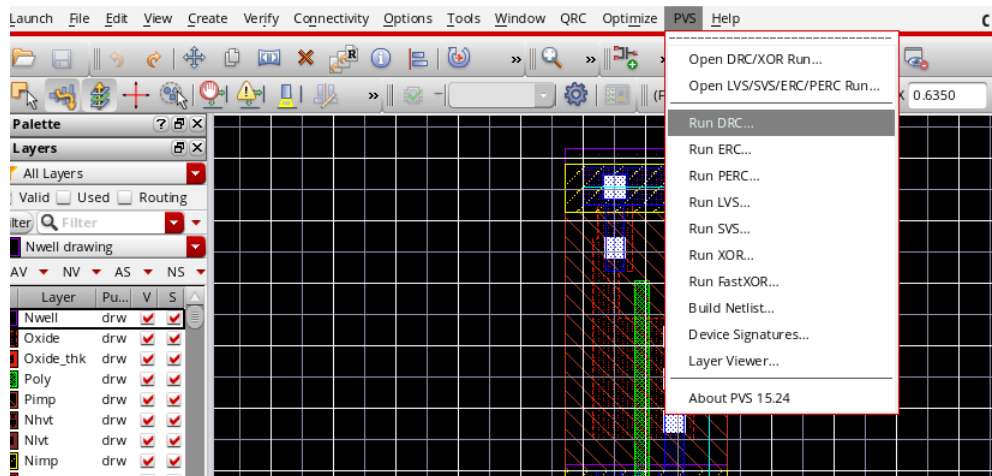


Figure 26: Run PVS DRC

A DRC setup form appears as shown in Figure 27.

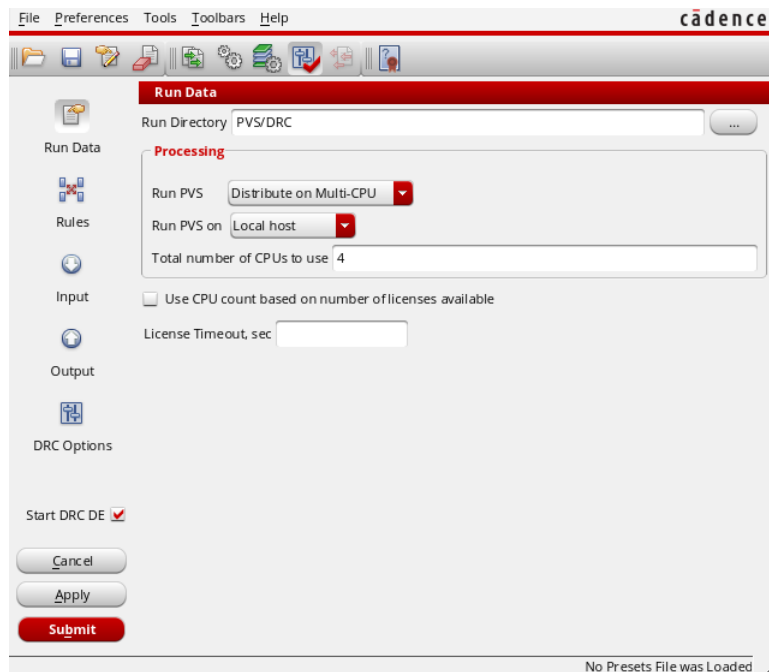


Figure 27: PVS DRC Setup Form - Configure Run Data

Set up the **Run Data** as shown in Figure 27; note that you are using the PVS/DRC directory you created earlier. Click on Rules.

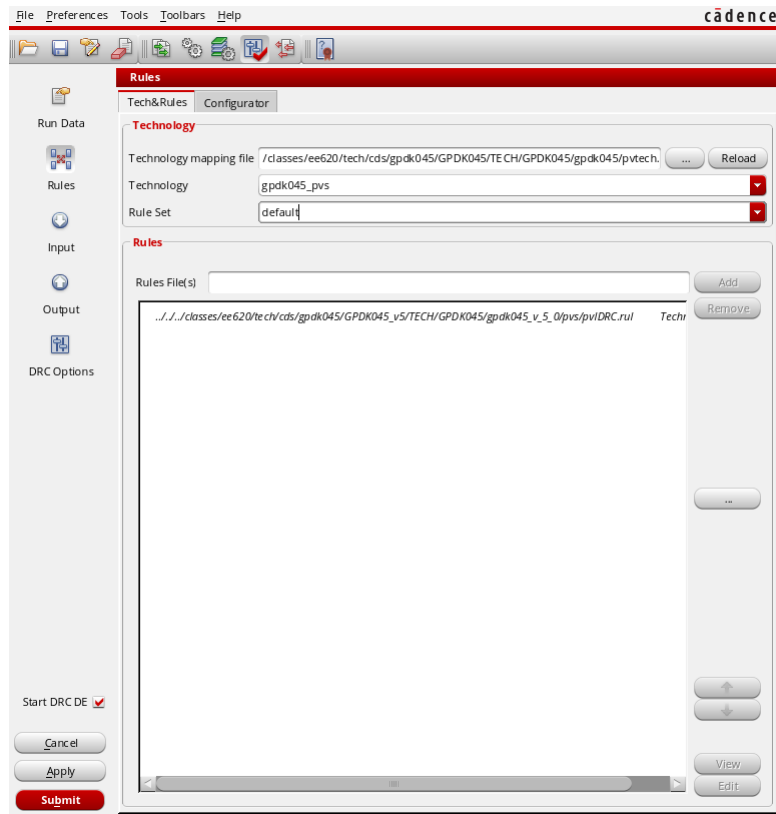


Figure 28: PVS DRC Setup form - Configure Rules

Set up the Rules as shown in Figure 28. Note that the technology mapping file is:

```
/classes/ee620/tech/cds/gpdk045/GPDK045/TECH/GPDK045/gpdk045/pvtech.lib
```

Click on **Input**, then configure it as shown in Figure 29.

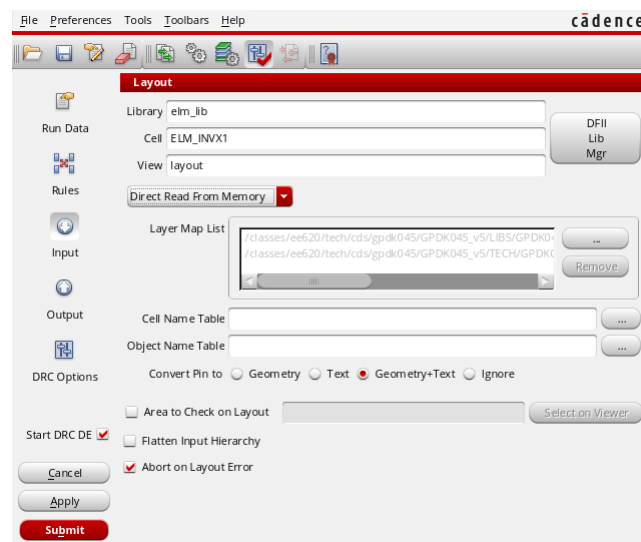


Figure 29: PVS DRC Setup form - Configure Input

Click on **Output**, then configure it as shown in Figure 30.

The screenshot shows the 'Output' configuration window in Cadence. The window has a sidebar on the left with icons for 'Run Data', 'Rules', 'Input', 'Output', and 'DRC Options'. The 'Output' icon is selected. The main area is titled 'Output' and contains the following settings:

- Report** section:
 - Name: ELM_INVX1.sum
 - Limit: 1000
 - Buttons: ☒ Replace File, ☐ Append To File, ☐ Statistics by Cell
- Output Format: ASCII (dropdown menu)
- Output File: PVS/DRC/ELM_INVX1.drc_errors.ascii (text field with browse button)
- Output Errors Hierarchically: Rules Definition (dropdown menu)
- ☐ Use Waivers for DRC Checks
- DRC DE** (radio button) and **Batch** (radio button) are selected.
- File(s): (empty text field with 'Add' and 'Remove' buttons)
- Start DRC DE: ☒ (checkbox)
- Add Suffix to Cells: (empty text field)
- Include Pseudo Cells: ☐ (checkbox)
- Vertices Per Polygon: 4096 (text field)
- Keep Layers: Rules Definition (dropdown menu)

At the bottom, there are buttons for 'Cancel', 'Apply', and 'Submit'.

Figure 30: PVS DRC Setup Form - Configure Output

Then click on **DRC Options**, and the DRC option menu should appear as shown in Figure 31.

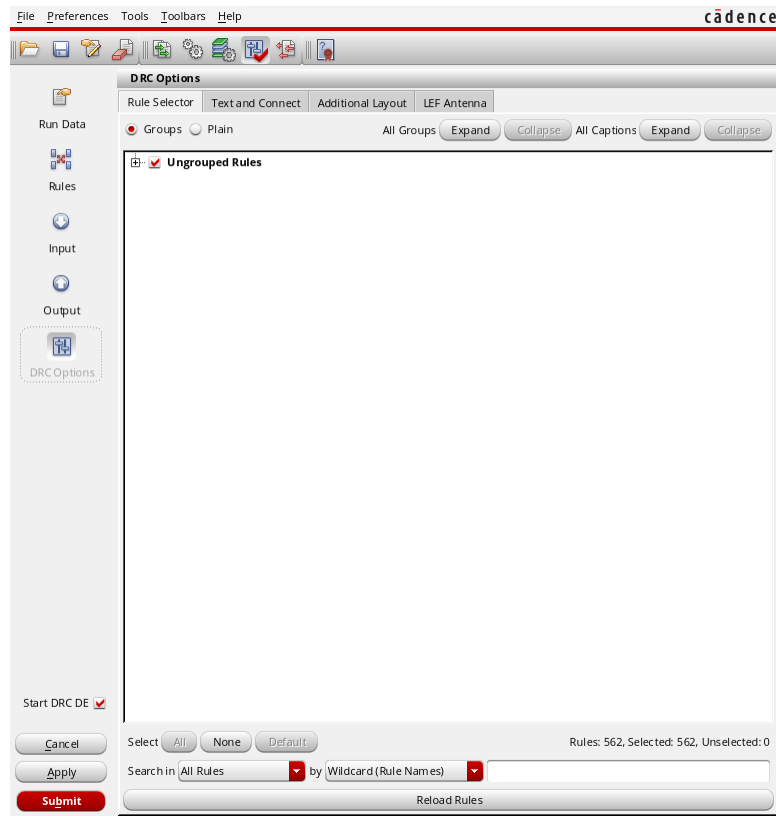


Figure 31: PVS DRC Setup Form - Configure DRC Options

Note that **you should NOT touch any of the DRC Options** as this can cause unintended effects. Before submitting, be sure to click the save button in the top left to save this setup to a file, as this will be used for all of your testing in the future.

Click on **Submit** to start the DRC job.

A few windows will appear on your screen. This includes a running log file and an error browser as shown below. **Note that this will look different because the tools have been updated, but the concepts still apply.**

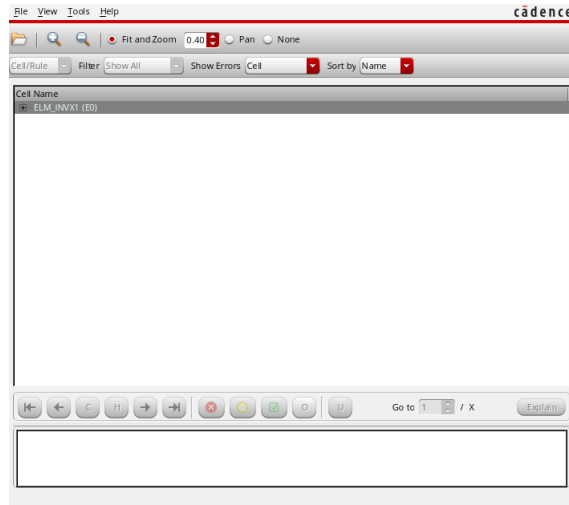


Figure 32: PVS DRC Error Browser

Figure 32 shows the DRC error browser. **E0** means my cell has been checked and zero errors have been identified. If errors are found, the Error Browser allows you to obtain details on the error (using the **Explain** button) and also to zoom to the error in your layout. Note that sometimes layout errors are reported several times because the issue might trigger a violation of several design rules. The PDK DRC rule book provides more detailed explanation on rules and how to go about fixing violations.

Please review the document GPK045_PDK_DESIGN_RULES.PDF posted in MyCourses for the 45nm PDK details and design rules. Fix the errors encountered and run DRC again. After you are done fixing the errors you can move on to the layout versus circuit schematic tests.

4.3 Run Layout Versus Schematic (LVS)

Select **PVS → Run LVS** (see Figure 33), LVS determines if the layout you created matches the schematic you created.

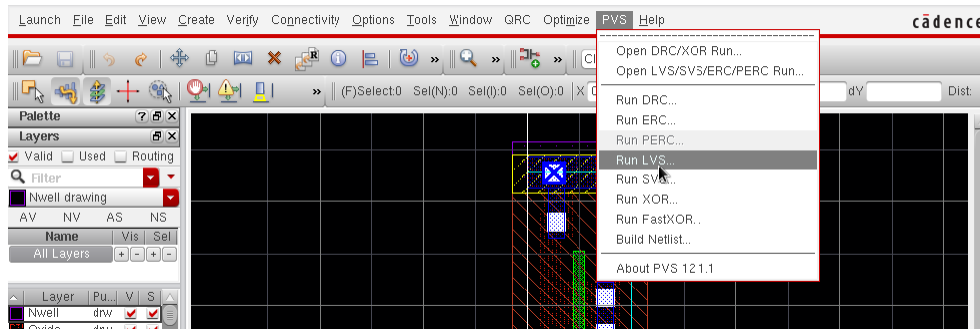


Figure 33: Run PVS LVS

A LVS form appears as shown in Figure 34.

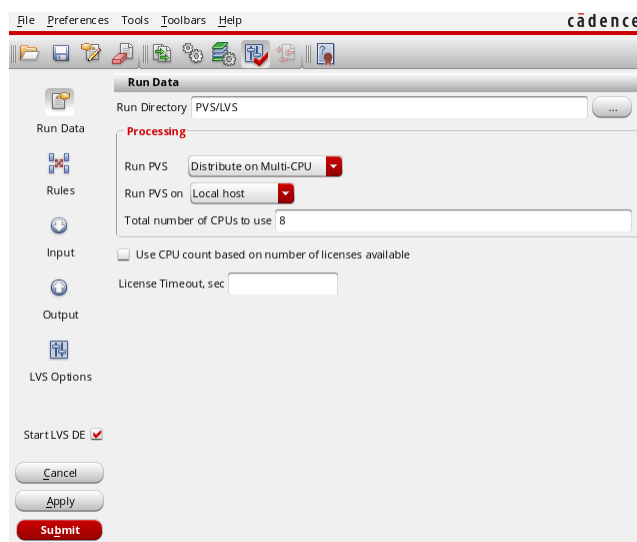


Figure 34: PVS LVS Setup Form - Configure Run Data

Setup the **Run Data** as shown in Figure 34; note that you are using the PVS/LVS directory you created earlier. Click on **Rules**, and the Rules configuration form (see Figure 35) should appear.

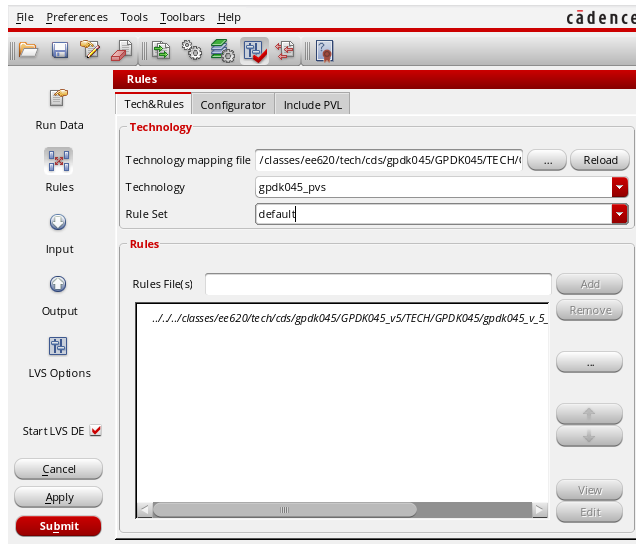


Figure 35: PVS LVS Setup Form - Configure Rules

Setup the **Rules** as shown in Figure 35. Note that the technology mapping file is:

`/classes/ee620/tech/cds/gpdK045/GPDK045/TECH/GPDK045/gpdK045/pvtech.lib`

Then click on **Input**

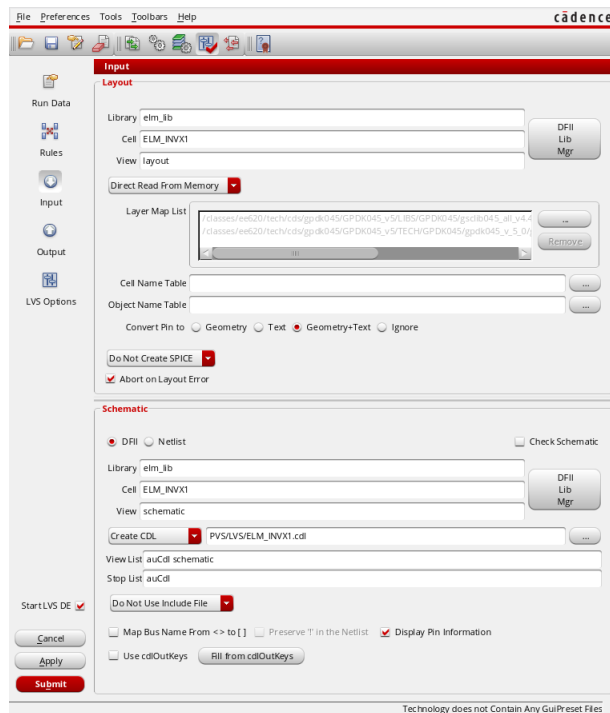


Figure 36: PVS LVS Setup Form - Configure Input

Setup the **Input** as shown in Figure 36.

Then click on **Output**.

The screenshot shows the 'Output' tab of the PVS LVS Setup Form. The 'Run ERC Checks' checkbox is checked. Under 'ERC Report', the 'Name' is 'ELM_INN1.sum', 'Limit' is '1000', and 'Output Format' is 'ASCII'. The 'Output Errors Hierarchically' checkbox is checked, and 'Rules Definition' is selected. Under 'LVS Report', the 'Name' is 'ELM_INN1.rep', 'Options' is '-none', and 'Limit' is '50'. The 'Mismatched Nets Limit' is '100'. Under 'Additional Output', the 'Create Quantus QRC Input Data' checkbox is checked, and 'QRCDataDir' is 'svdb'. The 'Keep Layers' dropdown is set to 'None'. The 'Start LVS DE' checkbox is checked. The 'Submit' button is highlighted in red.

Figure 37: PVS LVS Setup Form - Configure Output

Setup the Output as shown in Figure 37. Then click on **LVS Options**, then the LVS option window should appear (see Figure 38).

The screenshot shows the 'LVS Options' window. The 'ERC Rules and Groups' tab is selected. The 'Groups' radio button is selected. The 'All Groups' button is highlighted. The 'Select' dropdown is set to 'All'. The 'Search in' dropdown is set to 'All Rules'. The 'by' dropdown is set to 'Wildcard (Rule Names)'. The 'Reload Rules' button is highlighted. The 'Rules: 0, Selected: 0, Unselected: 0' status is shown. The 'Submit' button is highlighted in red.

Figure 38: PVS LVS Setup Form - Configure LVS Options

Note that **you should NOT touch any of the LVS Options** as this can cause adverse effects. Later in the course, you will need to add PVL options to ignore certain pad structures. Once again, make sure to click the save button in the top left before submitting so that you can save this preset for future testing.

Then, click on **Submit** to start the LVS job. A running log window will appear. If your schematic

and layout match you will get the window (see Figure 39):



Figure 39: PVS LVS Run Results Message Window

If there are mismatches, click yes and a LVS debug environment will be presented. Similar to the DRC debug environment, the LVS debugger enables cross probing between your layout and schematic and provides powerful ways to determine what differences are present. In this case, observe your layout (hint: most common errors have to do with the widths of the PMOS and NMOS and the proper connections to pins and intermediate nets)

4.4 Run Parasitic Resistor, Capacitor (RC) Extraction (RCX)

Assuming your LVS successfully passes without mismatches, select **QRC → Run PVS-Quantus QRC** as shown in Figure 40.

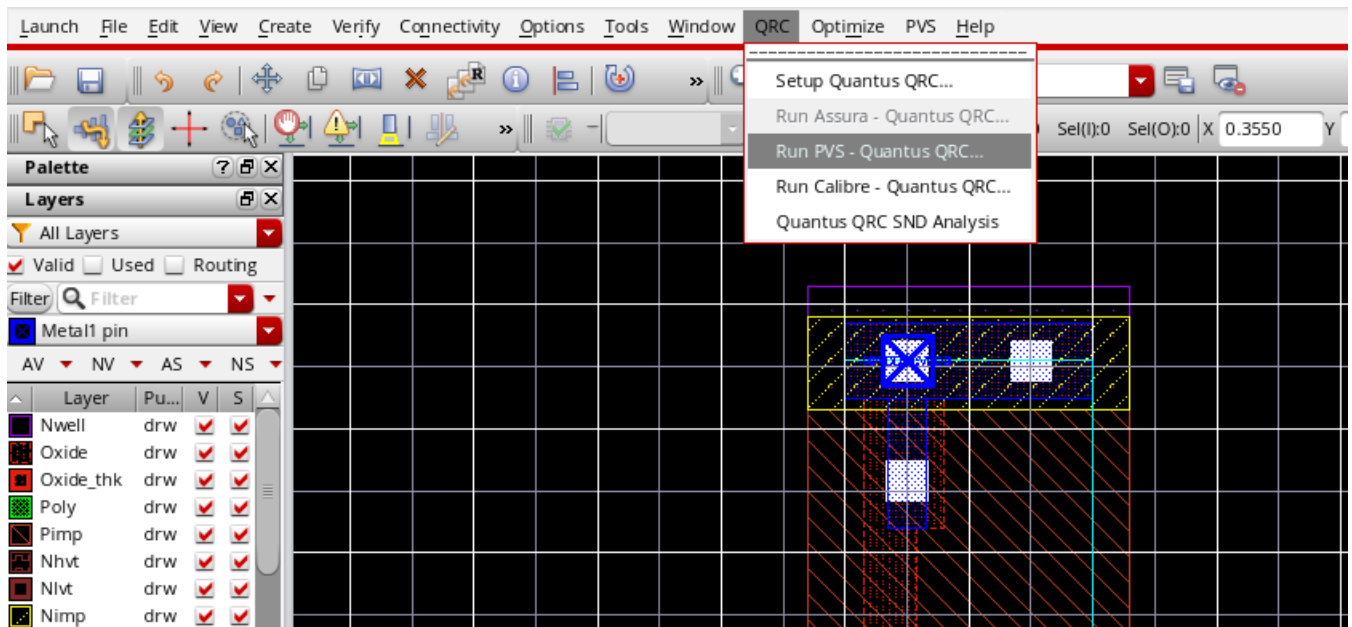


Figure 40: PVS-Quantus QRC Parasitic Extraction

The following window (see Figure 41) will appear:



Figure 41: PVS-QRC Setup Run Form

Confirm the settings and click OK. Then, A PVS-QRC settings window will appear as shown in Figure 42 and 43.

Setup Extraction Filtering Netlisting Run Details Substrate

Technology **gpdK045_pvs** RuleSet **rcx_typical**

☐ p2lvsSet **NONE** UseMultRuleSets ☐

☐ Setup Dir **ch/cds/gpdK045/GPDK045_v5/TECH/GPDK045/gpdK045_v_5_0/pvs**

☐ Include Command File

Rule Command File Include

☐ Tech Cmd File **User**

☐ Layer Setup File

☐ LPE Config File

☐ Library Cell Mapping File

☐ Probe Text File

Output **Extracted View** Lib **elm_lib** Cell **LM_INVX1** View **av_extracted**

Enable CellView Check ☒

Parasitic Res Component **presistor** Prop Id **r**

Parasitic Cap Component **pcapacitor** Prop Id **c**

Parasitic Ind Component **pinductor** Prop Id **l**

Parasitic M Component **pmind** Prop Id **k**

Inductance L1 Prop Id **ind1** Inductance L2 Prop Id **ind2**

Parasitic CCVS Component **ccvs** Hgain Prop Id **hgain** Vref Prop Id **vref**

Parasitic VS Component **vsouce** Prod Id **vr**

Library Directory: Specify a directory for writing local libraries created during the hierarchical extraction of an extracted view.

Figure 42: PVS-QRC Setup Form - Configure Setup Options (Top of Form)

The screenshot shows the 'Setup' tab of the PVS-QRC Setup Form. The form is organized into several sections with labels on the left and input fields on the right.

- Parasitic Cap Component:** Prop Id:
- Parasitic Ind Component:** Prop Id:
- Parasitic M Component:** Prop Id:
- Inductance L1 Prop Id:** **Inductance L2 Prop Id:**
- Parasitic CCVS Component:** Hgain Prop Id: Vref Prop Id:
- Parasitic VS Component:** Prod Id:
- Call Procedure:**
- Substrate Extract:** ☐ **Substrate Profile:**
- Add LVS MOS Diffusion Res:** ☐ **Extract MOS Diffusion Res:** ☐
- Auto Accuracy Downgrade:** ☒ **Extract MOS Diffusion High:**
- Library Prefix:**
- Library Directory:**
- Enable Import Color:** ☒ **DPT Corner Triplet RC Format:** ☐
- Custom DPT Corner File:**
- Gray Box:** ☐
- Pin Order File:**
- Force Subcircuit Pin Orders:** ☐
- CDL Out Run Dirs:**

At the bottom of the form, there is a message box:

Library Directory: Specify a directory for writing local libraries created during the hierarchical extraction of an extracted view.

Below the message box are several buttons:

Figure 43: PVS-QRC Setup Form - Configure Setup Options (Bottom of Form)

Adjust the Setup tab as shown in Figure 42 and 43, and then click on the **Extraction** tab, shown in Figure 44.

Setup **Extraction** Filtering Netlisting Run Details Substrate

Extraction Type **RC** Name Space **Layout Names**

Max fracture length **infinite** **microns** Temperature **25.0** C **Edit**

Max fracture length by layer **Edit**

☐ From File **...**

Cap Coupling Mode **Decoupled** Ref Node **VSS**

Mult Factor **1.0** Extend Checking Distance **default** Diffusion Equation R ☒

Ladder Network ☐

Select... User Region **...** **View** **Edit**

Net Selection Type **Full Chip All Nets** FS Extraction Mode **NONE**

Layer Setup Customization ☐ **Edit...** Quantus FS mode ☐

Field Solver High ☐

Field Solver Config File **...** **View** **Edit**

Resistance Mesh ☐ **Edit** Adaptive **false** **Mesh Via Layers**

Automatic Mesh Size ☐ R Mesh Accuracy **Moderate** **Options**

Auto Mesh Size Scale Factor **1.0** Error Tolerance **1.0** %

Select... R Mesh User Region **...** **View** **Edit**

Non-Manhattan Resistance Accuracy **default**

HRCX Cells: Specify a list of cells which appear in the output hierarchy, requires cell name, with optional view and lib names (cell, view, lib).

OK **Cancel** **Apply** **Defaults** **Load State** **Save State** **View Command File** **Help**

Figure 44: PVS - QRC Setup Form - Configure Extraction Options

Adjust the **Extraction** tab as shown in Figure 44, then click on the **Run Details** tab, shown in Figure 45.

Setup Extraction Filtering Netlisting **Run Details** Substrate

Run Name: ELM_INVX1

Run Directory: /home/e1m9478/CMPE630-digicTA/PVS/LVS/svdb

Log File: /home/e1m9478/CMPE630-digicTA/PVS/LVS/svdb/qrc.ELM_INVX1.log ... View

Keep Temporary Files ☐

Print Command File In Output Log ☐

Print Peak Memory In Output Log ☐

Quantus QRC Run Mode: Multi Processing ▼

Number of Processors: 4

Run Location: local ▼

Enable License Queue ☐ Timeout: 1800 seconds

Run Location: Submit Quantus QRC run for either local execution or remote execution on the specified machine name.

OK Cancel Apply Defaults Load State Save State View Command File Help

Figure 45: PVS - QRC Setup Form - Configure Run Details Options

Adjust the **Run Details** window as shown in Figure 45. Click OK to run the extraction.

Assuming the extraction completes successfully (see Figure 46), you will have an extracted view of your circuit with parasitic devices in your library.

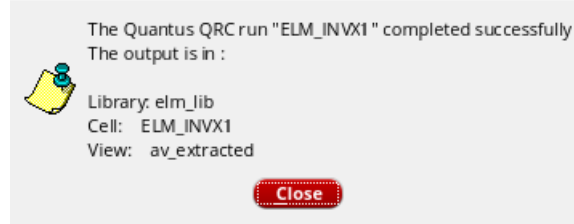


Figure 46: PVS - QRC Run Results Message Window

Using the Library Manager, open the view **av_extracted**. Make sure all layers are visible and you will see the extracted layout (ZOOM in to see the various parasitic devices, and as mentioned previously your extracted layout will look a lot different than the one shown in Figure 47).

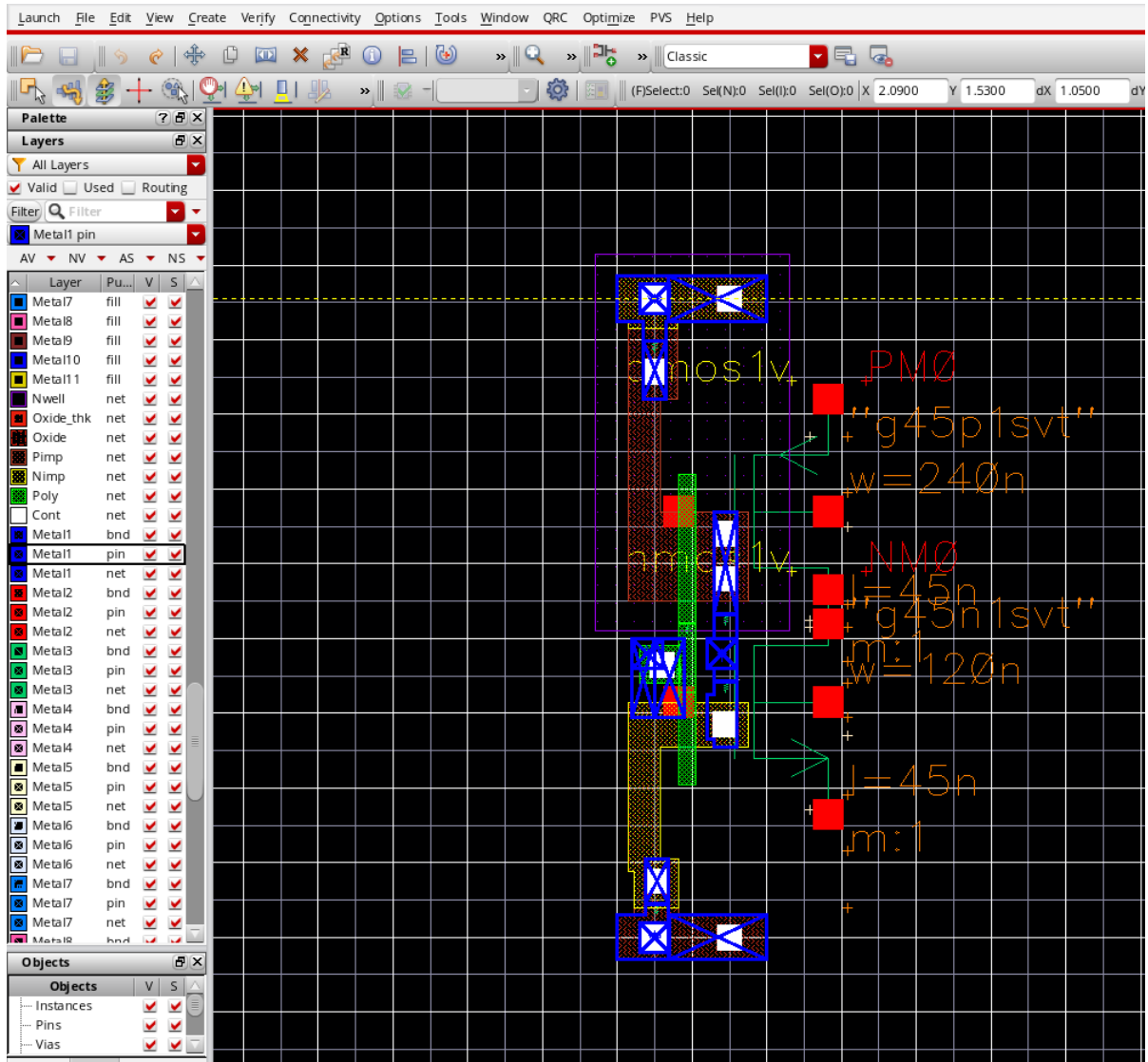


Figure 47: Extracted Parasitic Layout View

4.5 Simulate a Circuit (Post-Extraction)

Before you get started, close the Cadence tools and then log off. Log back in, return to your cmpe530/630 directory, and **restart** the Cadence tools..

Next, open the schematic cellview, (FML_INV_TEST).

We will now start the analog environment. Select **Launch** → **ADE-XL** from your testbench schematic. The following window (see Figure 48) will appear.

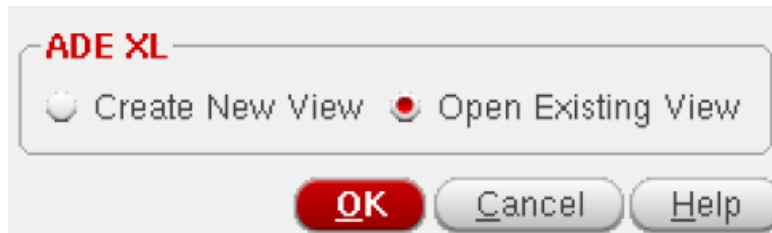


Figure 48: Launch ADE XL View Selection Form

Select **Open Existing View** and click OK. The open ADE XL window appears as shown in Figure 49.

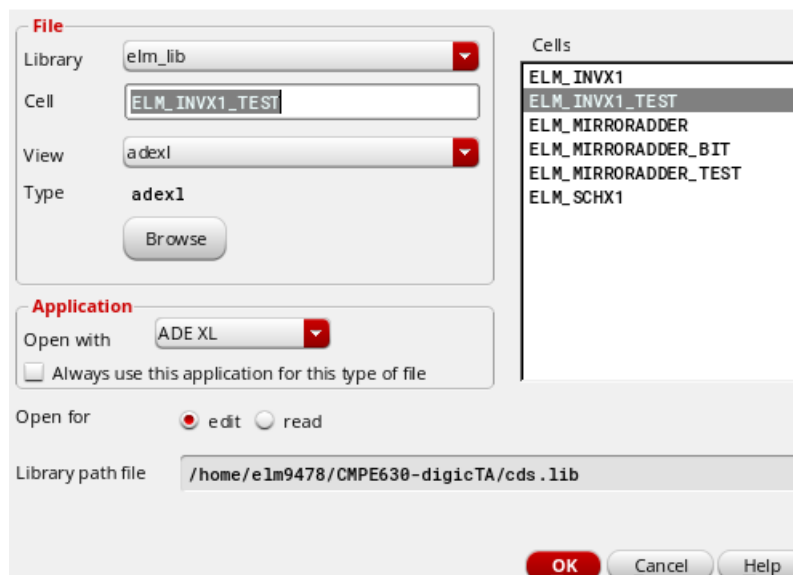


Figure 49: Launch ADE XL Cell Selection Form

Click OK. Then, the ADE XL window appears as a tab along with your testbench schematic as shown in Figure 50.

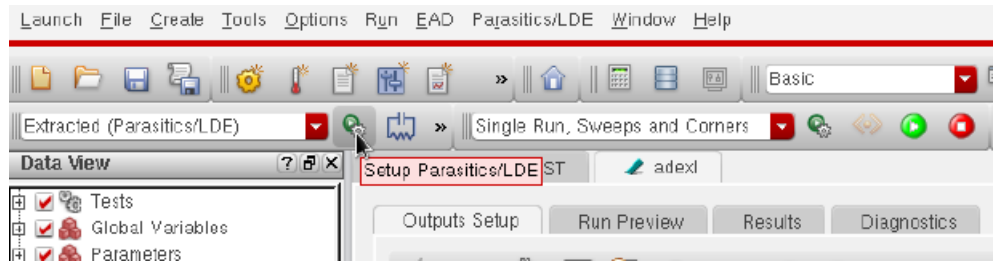


Figure 50: ADE XL Testbench Window

We will now setup the environment to use the extracted parasitic data during simulation. Click on the green gear to access the **Parasitics & LDE** setup from shown in Figure 51.

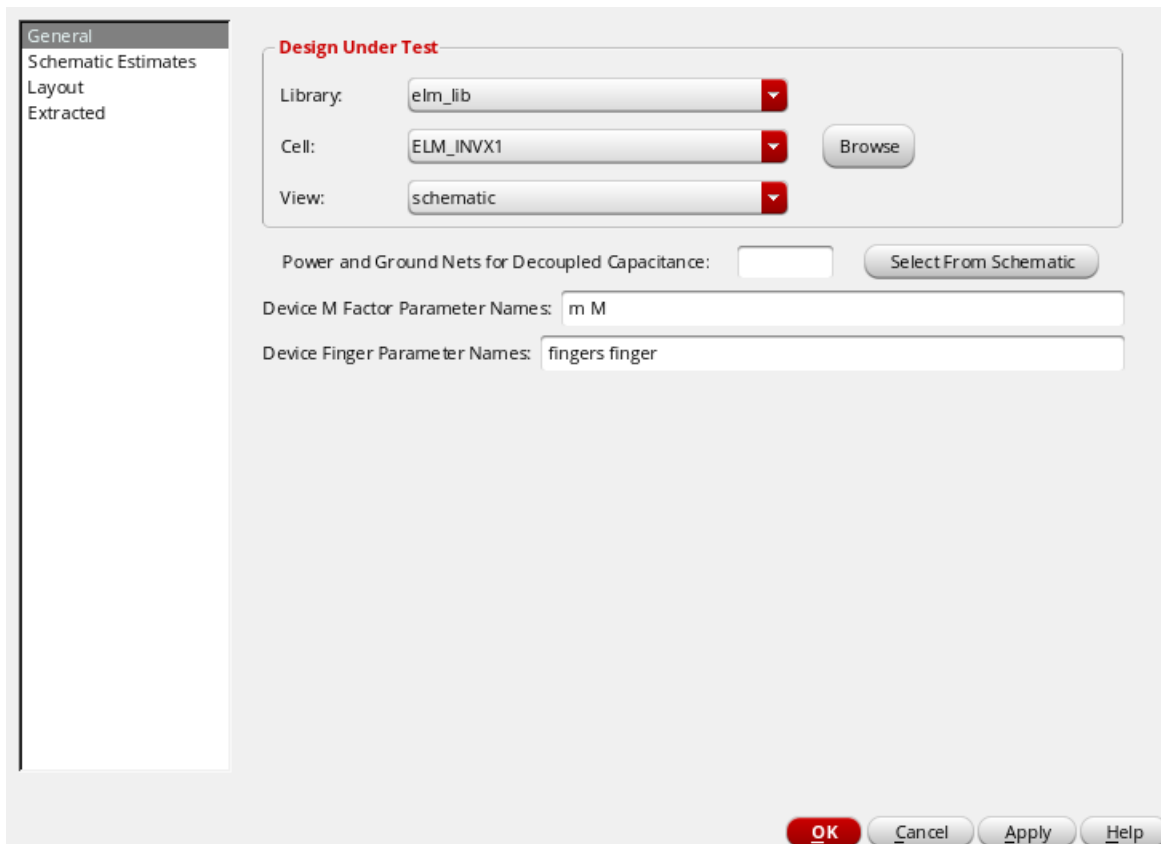


Figure 51: ADE XL Parasitic Setup - Configure General Settings

Setup the Design Under Test to be your Inverter cell as shown above. Then click on the Extracted tab as shown in Figure 52.

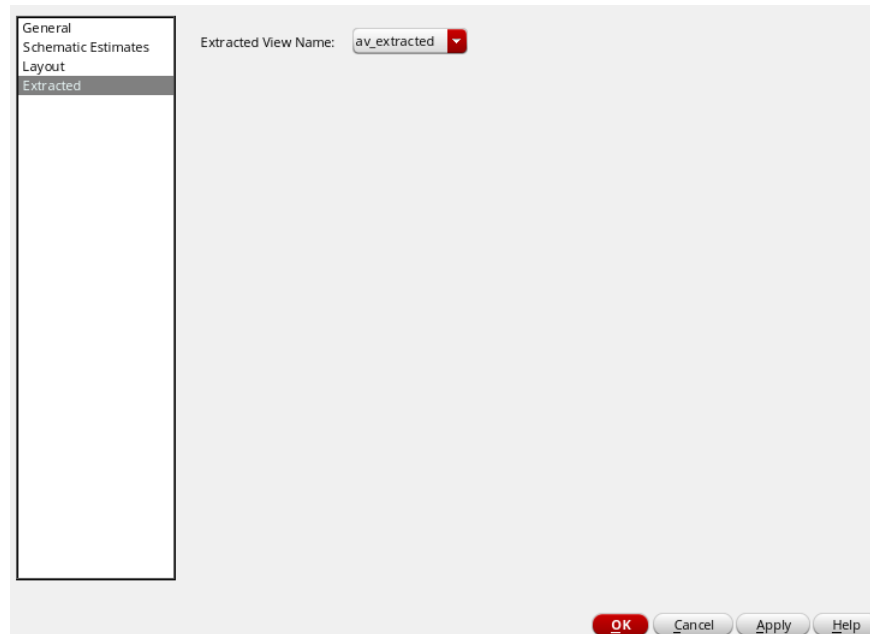


Figure 52: ADE XL Parasitic Setup - Configure Extracted Settings

Select the view **av_extracted** and click OK. Note that if the extracted-view-name view `av_extracted` does NOT appear as an option in the dropdown menu, select “Extracted” on the left, then click OK, and then open the Parasitics & LDE window again. This time, the drop down menu will show the correct extracted-view-name option.

You can now select the Extracted (Parasitics / LDE) mode as shown in Figure 53

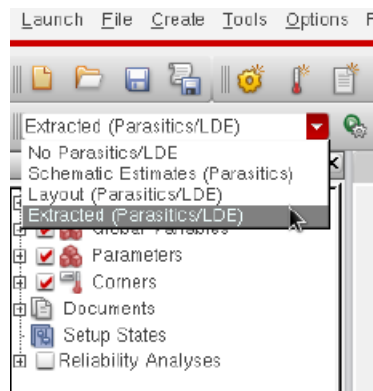


Figure 53: ADE XL Parasitic Setup

Rerun the simulations and observe the outputs. You should obtain similar results to your pre-layout simulation without any parasitic data. Measure the tests you created for Lab Exercise 2 and you should obtain slower results. This is because parasitics degrade circuit performance. Congratulations! You have successfully completed your first standard cell layout!

5 Results

Make sure to measure your design and screenshot it!

Table 1: Post-Extraction Inverter Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime	FallTime	T_P,HL	T_P,LH
Nominal	1.1	25	0				
	1.1	25	120				
Worst Case	1	125	0				
	1	125	120				

Table 2: Post-Extraction Overall Results

	VDD (V)	Temp (C)	Load (fF)	$F_{input,max}$	$F_{throughput,max}$
Inverter	1	125	0		
	1	125	120		

$$F_{input,max} = \frac{1}{t_{rise} + t_{fall}} \quad (1)$$

$$F_{throughput,max} = \frac{1}{T_{P,HL} + T_{P,LH}} \quad (2)$$

6 Questions

1. Using a screenshot of your av_extracted circuit. Examine and answer the following: What do each resistor and capacitor represent? Is it for poly, contact or what other material and how many are there?
2. Take the extracted view and draw the equivalent circuit with the parasitic capacitances and resistances.

7 Lab Exercise Report Requirements

The requirements for your lab report are described below. Please try your best to use the correct format for CE lab exercise reports that you have used in the past. Please follow the guidelines and also apply as much as you can from what you already know to explain theory, waveforms, results, circuit behavior, and everything else in a proper report!

Note: Proper citation is needed for all non-original figures (i.e., those that you have not made yourself).

1. Cover Page

- Name and number of the lab exercise
- Your name, name of the instructor, and name of the TAs
- Submission Date

2. Table of contents

3. Abstract

- A description of what you did for this exercise, how you did it, and significant numerical lab results

4. Design Methodology

- Include all necessary figures such as inverter schematic, stick diagram, and layout
 - Stick diagram must be made and labeled with different colors
- Discuss each layout layer: What materials are used and why?
- Report the height, width, and area of your layout
- Briefly Discuss DRC, LVS, and QRC. A simple description of what they are will suffice

5. Result and Analysis

- Include clean DRC and LVS output message
- Include a capture of your **Extracted Parasitic Layout**
 - Report the number of NMOS and PMOS transistors used in your inverter cell
 - Report the number of resistors and capacitors in your Extracted Parasitic Layout
- Include the following simulation results:
 - Transient simulation waveform of the inverter (from Lab 2)
 - Pre-extraction test results (same as Lab 2) in the ADE XL editor
 - Post-extraction test results (from this lab) in the ADE XL editor
- Complete and include Table 1 and 2 (include units)
- Compare the timing simulations of your extracted netlist with those from Lab 2
 - Are the timing faster or slower for the post-extraction simulation? Explain why the timing is different.
- Calculate and report the percentage increase or decrease in the maximum operating frequency and the maximum throughput between the worst case with and without load

- Calculate and report the percentage of how fast/slow the post-extraction inverter is compared to the pre-extraction inverter
 - Calculate and report the speedup/slowdown of the post-extraction inverter **worst case loaded** using the maximum operating frequency and throughput

6. Conclusions

- Restate important results, how they were obtained, any uncertainties, possible improvements, and any implications
- This should not be a copy-paste of the Abstract

7. Questions

- Re-write the question(s)
- Answer all the questions from the "questions" section

8. Sign-off Sheet

- **This is absolutely required! Lab reports will not be graded without a sign-off sheet, resulting in an automatic zero.**

8 Signoff Sheet

Table 3: Inverter Layout

Students Name:

Section:

Prelab	Point Value	Points Earned	Date
Inverter Stick Diagram	10		
Demo	Point Value	Points Earned	Date
DRC check	25		
LVS check	25		
Parasitic RC RCX	20		
Post-Extraction Results	20		
Total Points	100		